
When Softmax Fails at the Top: Extreme-Value Corrections for InfoNCE

Melihcan Erol^{*1} Suat Evren^{*1} Oktay Ozel² Alexander Morgan¹ Jongha Jon Ryu¹ Lizhong Zheng¹

Abstract

InfoNCE is a standard loss in contrastive learning, but its softmax form is not only a computational convenience; it also encodes a statistical assumption about how the top-scoring example is selected. Using extreme value theory, we show that this assumption is often misaligned with the normalized embedding setting used in modern contrastive learning. Motivated by this, we propose WEINCE, a simple modification of InfoNCE that uses anchor-wise online batch statistics to blend the usual softmax logits with an endpoint shortfall correction, adding no trainable parameters. Across five vision benchmarks, including Tiny-ImageNet, and on SimCSE sentence embeddings, WEINCE yields consistent improvements in frozen-feature evaluation. These results show that a more faithful statistical treatment of hard negatives can improve contrastive objectives.¹

1. Introduction

Contrastive learning has become a central approach to self-supervised representation learning, and InfoNCE or closely related softmax-based objectives sit at the core of many influential methods in vision, language, and multimodal learning (Jaiswal et al., 2021; Le-Khac et al., 2020; van den Oord et al., 2018; Chen et al., 2020; He et al., 2020; Gao et al., 2022; Radford et al., 2021). The standard interpretation is that the model learns to assign high similarity to related views of the same example and low similarity to views from different examples (van den Oord et al., 2018; Chen et al., 2020; He et al., 2020). This approach has proved highly effective for learning transferable representations from unlabeled data (Jaiswal et al., 2021; Le-Khac et al., 2020; Chen et al., 2020; He et al., 2020; Gao et al., 2022; Radford et al.,

2021).

In many of these systems, similarities are computed after normalizing the learned representations, typically through cosine similarity or an equivalent scaled inner product on the unit sphere (Chen et al., 2020; Wang & Isola, 2020; Gao et al., 2022; Radford et al., 2021). This places the loss in a bounded similarity regime. At the same time, prior work shows that contrastive learning is highly sensitive to the negative set. SimCLR reports strong gains from larger batch sizes and longer training (Chen et al., 2020), theoretical analyses emphasize the role of the number of negatives in representation quality (Saunshi et al., 2019), and several works show that hard negatives receive disproportionate weight under contrastive losses (Wang & Liu, 2021; Kalantidis et al., 2020; Robinson et al., 2021). Taken together, these observations suggest that the high-similarity tail of the negative distribution deserves special attention.

In most of this literature, the softmax in InfoNCE appears simply as part of the loss definition. Contrastive learning and InfoNCE are often justified through information-theoretic arguments, for example by showing that minimizing the loss maximizes a lower bound on mutual information (van den Oord et al., 2018; Chen et al., 2020; Lu et al., 2023; Tian et al., 2020; Poole et al., 2019; Alshammari et al., 2025). In this paper, we show that the softmax is doing something more specific. Discrete choice theory shows that softmax probabilities arise from the Plackett–Luce model for top-1 selection (Luce, 1959; Plackett, 1975). This provides a new perspective on InfoNCE: when the positive example wins against a set of negatives, the loss is implicitly fitting a statistical model for that winning event. This viewpoint lets us ask whether the hidden assumption behind softmax is compatible with the normalized embedding regime used in modern contrastive learning.

We answer this question negatively in the regime that matters most. The softmax assumption is not well aligned with normalized embeddings near the top of the similarity range, where the hardest negatives live. This leads to WEINCE (Weibull-Enhanced InfoNCE), a simple modification of InfoNCE that uses online batch statistics to interpolate between standard softmax logits and an endpoint shortfall correction. The interpolation is deliberate: it recovers InfoNCE when anchor-wise evidence for endpoint behavior is

^{*}Equal contribution ¹Massachusetts Institute of Technology, Cambridge, Massachusetts ²Boston University, Boston, Massachusetts. Correspondence to: Melihcan Erol <hsmerol@mit.edu>, Suat Evren <evrenis@mit.edu>.

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¹Code will be released at <https://github.com/hsme98/weince>.

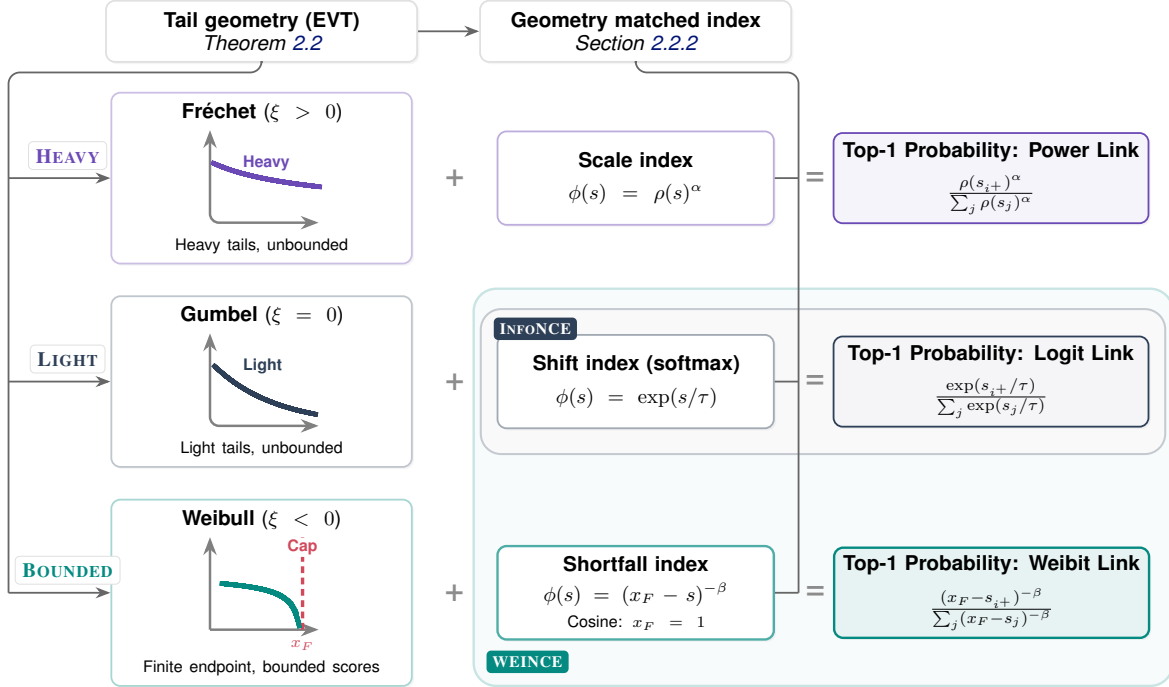


Figure 1. Extreme value theory (Theorem 2.2) yields three tail geometries. Pairing each geometry with a compatible index model gives a corresponding top-1 score. The Gumbel row recovers the softmax score used by InfoNCE. For bounded similarities, the Weibull row motivates a shortfall-based logit. WEINCE combines the softmax and shortfall logits through interpolation, so InfoNCE is recovered as the special case $\lambda = 0$.

weak and moves toward the Weibull shortfall link only when the current negative set is near the score cap. We support this view with theory, diagnostics, and experiments, and we show consistent improvements in frozen-feature evaluation across five vision benchmarks and SimCSE sentence embeddings.

Our contributions are threefold. First, we make explicit the statistical assumption hidden inside the standard InfoNCE softmax. Second, we show why this assumption becomes problematic in normalized embedding spaces and how the mismatch appears in likelihood and gradient-allocation diagnostics. Third, we introduce WEINCE, a practical replacement for InfoNCE that uses anchor-wise tail evidence to interpolate between softmax and Weibull shortfall logits, rather than assuming a pure Weibull correction for every anchor.

2. Revisiting InfoNCE via Extreme-Value Theory

Self-supervised representation learning aims to learn an encoder f from *unlabeled* data so that its representations transfer well to supervised downstream tasks, especially when labeled data is scarce. A standard pipeline first pre-trains an encoder backbone with a self-supervised objective and then either fine-tunes the encoder or trains a lightweight

prediction head using the available labels.

Contrastive learning implements self-supervision by constructing multiple views of the same underlying data point and treating these views as a positive pair, while treating views from other data points as negatives. A widely used contrastive objective in this setting is the InfoNCE loss.

Definition 2.1 (InfoNCE loss). Let $(X, Y_0) \sim p_{XY}$ denote two correlated views (for example, an input and an augmentation), and let $Y_1, \dots, Y_K \stackrel{\text{i.i.d.}}{\sim} p_X$ be negatives drawn independently of (X, Y_0) . Write $U = f(X)$ and $V_i = f(Y_i)$ for $i \in \{0, 1, \dots, K\}$. The population InfoNCE loss is

$$\mathcal{L}(f) \triangleq -\mathbb{E} \left[\log \frac{\exp(s(U, V_0)/\tau)}{\sum_{i=0}^K \exp(s(U, V_i)/\tau)} \right], \quad (2.1)$$

with empirical counterpart

$$\hat{\mathcal{L}}(f) \triangleq -\frac{1}{m} \sum_{j=1}^m \log \frac{\exp(s(u_j, v_{j0})/\tau)}{\sum_{i=0}^K \exp(s(u_j, v_{ji})/\tau)}. \quad (2.2)$$

where $s(\cdot, \cdot)$ is cosine similarity

$$s(u, v) = \frac{\langle u, v \rangle}{\|u\|_2 \|v\|_2},$$

and $\tau > 0$ is the temperature.

2.1. InfoNCE as a Preference Model Estimation

Objective

Each term in Equation (2.2) can be viewed as the negative log-likelihood of a *top-1 choice* event: for an anchor view u_j , the positive v_{j0} is observed to “win” among the candidate set $\{v_{ji}\}_{i=0}^K$ (positive plus K negatives). This connects InfoNCE to learning-to-rank and discrete choice models.²

A general random-utility generative model. Fix an anchor $j \in \{1, \dots, m\}$ and let $\{s_{ji}\}_{i=0}^K$ denote the (latent) scores assigned to the $K+1$ candidates. A broad way to model top-1 outcomes is through a *random utility model* in which each candidate i is associated with a latent utility U_{ji} , drawn independently across candidates from a conditional family

$$U_{ji} \mid s_{ji} \sim P_{U|S=s_{ji}}, \quad i \in \{0, 1, \dots, K\},$$

and the observed outcome reveals only the winner

$$I_j = \arg \max_{0 \leq i \leq K} U_{ji}.$$

Under this lens, a contrastive dataset provides outcomes $\mathcal{I} \triangleq \{I_j\}_{j=1}^m$, and the learning problem is to parameterize the scores via an encoder and maximize the likelihood of the observed winners. Since the candidate indices are arbitrary up to relabeling, we may, without loss of generality, index the positive candidate as 0 for every anchor.

Recovering InfoNCE as a special case. InfoNCE corresponds to a particular *parametric* choice of the conditional model $P_{U|S}$, namely an *additive* random-utility model with i.i.d. Gumbel noise:

$$U_{ji} = s_{ji} + \epsilon_{ji}, \quad \epsilon_{ji} \stackrel{\text{i.i.d.}}{\sim} \text{Gumbel}(0, \tau). \quad (2.3)$$

Under Equation (2.3), the top-1 probability takes the softmax form

$$\mathbb{P}(I_j = k \mid \{s_{ji}\}_{i=0}^K) = \frac{\exp(s_{jk}/\tau)}{\sum_{i=0}^K \exp(s_{ji}/\tau)}.$$

We parameterize scores using an encoder $f : \mathcal{V} \rightarrow \mathbb{R}^d$ via cosine similarity,

$$s_{ji} = \frac{\langle f(u_j), f(v_{ji}) \rangle}{\|f(u_j)\|_2 \|f(v_{ji})\|_2},$$

so the log-likelihood of observing $\mathcal{I}$ becomes

$$\begin{aligned} \ell(f) &\triangleq \log \mathbb{P}(\mathcal{I}) \\ &= \sum_{j=1}^m \log \mathbb{P}(I_j = 0) \\ &= \sum_{j=1}^m \log \frac{\exp(s(u_j, v_{j0})/\tau)}{\sum_{i=0}^K \exp(s(u_j, v_{ji})/\tau)}. \end{aligned} \quad (2.4)$$

²In the classical parametric case this reduces to the Plackett–Luce (multinomial logit) model.

Up to the factor $1/m$, the negative log-likelihood $-\ell(f)$ is exactly the empirical InfoNCE objective in Equation (2.2).

Implicit assumptions behind the softmax link. This likelihood view highlights what InfoNCE implicitly assumes about how top-1 outcomes are generated. The softmax form is not only tied to the *Gumbel* distributional choice, but also to the *additive-in-score* structure in Equation (2.3). More generally, however, the latent “utility” driving which candidate wins need not be an additive perturbation of the score: one could instead posit an unknown conditional model $P_{U|S}$ (potentially non-additive, heteroskedastic, or operating in a different tail coordinate altogether). This leads to the guiding question of the paper:

When is the softmax top-1 likelihood implied by Equation (2.3) a good approximation to the top-1 outcomes generated by an unknown conditional utility model $P_{U|S}$, and what replaces it when it is not?

At this level of generality, the question is underdetermined: there are many ways to specify $P_{U|S}$, and different choices induce different top-1 link functions. To obtain principled guidance, we turn to asymptotics.³ In modern contrastive learning, K can be large, and the top-1 event is governed by the extreme behavior of $\max_i U_{ji}$, or equivalently by the tail behavior of the utility family. Just as the CLT provides a universal limit theory for sums, extreme value theory (EVT) provides a universal limit theory for maxima. This motivates our next step: analyze the $K \rightarrow \infty$ regime using EVT to identify which aspects of $P_{U|S}$ matter for top-1 probabilities and to derive geometry-consistent alternatives to the softmax link.

2.2. Understanding Large- K Behavior of InfoNCE via Extreme Value Theory

2.2.1. A TAIL GEOMETRY TRICHOTOMY

We now turn to extreme value theory (EVT), which characterizes the distributional behavior of maxima. The foundational result is the Fisher–Tippett–Gnedenko theorem (Gnedenko, 1943; de Haan & Ferreira, 2006), which can be understood as a central limit theorem for maxima.

Theorem 2.2 (Fisher–Tippett–Gnedenko). *Let $X_1, X_2, \dots$ be i.i.d. random variables with common CDF F , and let $M_n = \max_{1 \leq i \leq n} X_i$. If there exist normalizing sequences*

³A recurring theme in statistics is that large-sample or high-dimensional limits reveal *universal* structure. Classical inference for sums rests on the central limit theorem (e.g., (van der Vaart, 1998)), while the Tracy–Widom law is a canonical universality result for the fluctuations of extreme eigenvalues of large random matrices (e.g., (Tracy & Widom, 1994; Anderson et al., 2010; Johnstone, 2001)).

$a_n > 0$ and $b_n \in \mathbb{R}$ such that

$$\mathbb{P}\left(\frac{M_n - b_n}{a_n} \leq x\right) \rightarrow G(x)$$

for a nondegenerate CDF G , then G is of generalized extreme value type. Thus there are $\mu \in \mathbb{R}$, $\sigma > 0$, and $\xi \in \mathbb{R}$ such that

$$G(x) = G_\xi\left(\frac{x - \mu}{\sigma}\right), \quad G_\xi(z) = \exp\{-\nu_\xi(z)\},$$

where

$$\nu_\xi(z) = \begin{cases} \exp(-z), & z \in \mathbb{R}, \quad \xi = 0, \\ (1 + \xi z)^{-1/\xi}, & 1 + \xi z > 0, \quad \xi \neq 0. \end{cases}$$

Equivalently, after the canonical restandardization $\tilde{b}_n = b_n + \mu a_n$ and $\tilde{a}_n = \sigma a_n$, on compact subsets of the interior support,

$$\frac{\tilde{F}(\tilde{b}_n + \tilde{a}_n z)}{\tilde{F}(\tilde{b}_n)} \rightarrow \nu_\xi(z), \quad \tilde{F} = 1 - F. \quad (2.5)$$

This single parameter ξ describes three qualitatively different tail geometries (see Figure 2):

1. **Fréchet** ($\xi > 0$): heavy tails (power law type) with unbounded right tail.
2. **Gumbel** ($\xi = 0$): rapidly or exponentially decaying tails under the appropriate normalization; the limiting law has support on all $\mathbb{R}$.
3. **Weibull** ($\xi < 0$): finite right endpoint with regular variation in the shortfall to that endpoint; after affine normalization the support is bounded above.

Strictly speaking, a domain of attraction is a property of a tail law, not of one realized finite set of scores. Moreover, bounded support alone does not force the Weibull domain: finite-endpoint distributions with sufficiently rapid decay near the endpoint can still be Gumbel-type. Thus, for bounded cosine similarities, the relevant question is local and anchor-dependent: whether the active hard-negative tail behaves more like a translation/Gumbel tail or like an endpoint shortfall/Weibull tail.

Remark 2.3. For the original normalizers, the invariant tail-ratio statement is

$$\frac{\tilde{F}(b_n + a_n x)}{\tilde{F}(b_n + a_n x_0)} \rightarrow \frac{-\log G(x)}{-\log G(x_0)}$$

for any reference point x_0 with $0 < G(x_0) < 1$. Equation (2.5) is the canonical form obtained after choosing location and scale so that the limiting CDF is the standard G_ξ and $x_0 = 0$.

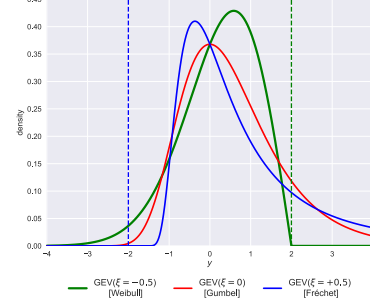


Figure 2. Examples of distributions exhibiting the three tail geometries (Fréchet, Gumbel, Weibull) that arise as limits for normalized maxima under Theorem 2.2.

An extension to nonidentical utilities. Conditional on their realized scores, candidate utilities are independent but generally nonidentically distributed. Appendix A proves a permutation block extension of FTG under infinitesimality and random block representativeness: a uniformly selected block of n candidates from an mn candidate reservoir has the same n scale extreme limit as the original pool. The same appendix also gives a short conditional FTG alternative under the i.i.d. score indexing in Definition 2.1; exchangeability explains why the block condition is natural for contrastive candidate sampling.

2.2.2. CANONICAL INDEX MODELS GENERATING A TAIL GEOMETRY

Section 2.2.1 shows that, in the large candidate regime, maxima exhibit one of three universal tail geometries (Fréchet, Gumbel, or Weibull). This suggests a modeling principle that we use throughout the paper: rather than committing a priori to a specific parametric form such as additive Gumbel noise as in Equation (2.3), we can first ask which tail geometry the latent utilities plausibly follow and then use a ranking model that is compatible with that geometry. Equivalently, each admissible tail geometry can be paired with a canonical random utility model whose induced maxima lie in that geometry. Since the observed label is a top-1 outcome, selecting a model with mismatched effective geometry can distort the relative probabilities in the extreme tail and can be statistically inefficient for estimating the underlying scores from winner observations⁴.

These canonical ranking models can be viewed as generalizations of the Plackett–Luce model in Equation (2.3), obtained by working in an appropriate tail coordinate. Concretely, there exists a monotone transform T and an index

⁴This is the standard quasi maximum likelihood interpretation formalized in Theorem C.1; see also (White, 1982; Gouriéroux et al., 1984; van der Vaart, 1998).

Table 1. Tail geometry (ξ) and extreme value invariance. A geometry matched indexing becomes additive in the tail coordinate: $T(X_s) = \eta(s) + \varepsilon$ with candidate independent noise ε (for example, $\varepsilon = \epsilon$ in the Gumbel row, $\varepsilon = \log Z$ in the Fréchet row, and $\varepsilon = -\log W$ in the Weibull row). Here $\alpha = 1/\xi > 0$ in the Fréchet case and $\beta = -1/\xi > 0$ in the Weibull case.

Geometry	Natural action	Tail coordinate $T(x)$	Index statistic $\eta(s)$	Canonical model	Top 1 rule
$\xi = 0$ (Gumbel)	translation	$T(x) = x$	$\eta(s) = s$	$X_s = s + \epsilon$	$p_i \propto \exp(s_i/a_n)$
$\xi > 0$ (Fréchet)	scaling	$T(x) = \log x$	$\eta(s) = \log \rho(s)$	$X_s = \rho(s) Z$	$p_i \propto \rho(s_i)^\alpha$
$\xi < 0$ (Weibull)	endpoint shortfall	$T(x) = -\log(b - x)$	$\eta(s) = -\log q(s)$	$b - X_s = q(s) W$	$p_i \propto q(s_i)^{-\beta}$

map $\eta(\cdot)$ such that utilities satisfy

$$T(X_i) = \eta(s_i) + \varepsilon_i, \quad (2.6)$$

where ε_i is candidate independent noise. Inverting the tail coordinate yields three representative forms:

- (a) **Multiplicative utility (Fréchet):** $X_i = \rho(s_i) Z_i$.
- (b) **Additive utility (Gumbel):** $X_i = s_i + \epsilon_i$.
- (c) **Shortfall to endpoint (Weibull):** $X_i = b - q(s_i) W_i$.

Here Z_i, ϵ_i, W_i are i.i.d. noise variables whose tails lie in the Fréchet, Gumbel, and Weibull domains, respectively.

2.2.3. ASYMPTOTICALLY CORRECT LINK FUNCTIONS

Recall the general formulation $U \mid S = s \sim P_{U|S=s}$ introduced earlier. Let F_s be the CDF of $U \mid S = s$ and $\bar{F}_s(t) \triangleq \mathbb{P}(U > t \mid S = s)$ its survival function. The next result shows that, for large candidate sets, the top-1 selection probabilities depend on $P_{U|S}$ only through these tail probabilities evaluated at the normalization level b_n from Theorem A.2. The assumptions and proof are in Appendix B.

Theorem 2.4 (Tail mass proportionality). *Let $b_n \in \mathbb{R}$ and $a_n > 0$ be as in Theorem A.2. Then under Assumption B.1, uniformly in $1 \leq i \leq n$,*

$$p_i \triangleq \mathbb{P}(I_n = i \mid \mathbf{s}_n) = \frac{\bar{F}_{s_i}(b_n)}{\sum_{r=1}^n \bar{F}_{s_r}(b_n)} (1 + o(1)).$$

Theorem 2.4 reduces the large K top-1 link induced by $P_{U|S}$ to the tail weights $\bar{F}_{s_i}(b_n) = \mathbb{P}(U > b_n \mid S = s_i)$. The FTG tail ratio characterization (Theorem 2.2) implies that, within a fixed EVT domain of attraction (equivalently, for a fixed tail geometry parameter ξ), these tail weights have universal asymptotic ratios. In Theorem B.2 we make this dependence on ξ explicit, yielding geometry-dependent top-1 links. Evaluating the resulting tail weights for the canonical models in Section 2.2.2 gives the closed form probabilities in Table 1. The table should be read conditionally: different anchors, training stages, or negative pools may exhibit different effective tail geometries, and a finite

minibatch cannot identify the asymptotic domain with certainty. Thus the theory motivates an adaptive link, rather than a global replacement of softmax by a single alternative. In the next section, we test whether winner outcomes observed during InfoNCE training show systematic endpoint evidence beyond the softmax link implied by Equation (2.3).

Shape correctness in the Weibull branch. The Weibull rule in Table 1 can be written as a softmax after changing coordinates:

$$\frac{q(s_i)^{-\beta}}{\sum_j q(s_j)^{-\beta}} = \text{softmax}_i(-\beta \log q(s_i)).$$

Thus the issue is not normalized exponentiation itself, but the standard InfoNCE use of an affine raw-score logit s/τ . By Lemma C.2, the endpoint link agrees with raw-score InfoNCE on all candidate sets only if $-\beta \log q(s)$ is affine in s . In the natural local model $q(s) = c(x_F - s)^\kappa (1 + o(1))$, the correct endpoint coordinate is $-\beta\kappa \log(x_F - s) + O(1)$, which cannot be recovered by retuning a single temperature.

2.3. InfoNCE May Be Misspecified for Contrastive Representation Learning

Recall from Equation (2.3) that InfoNCE coincides with maximum likelihood when utilities follow an additive random utility model with i.i.d. Gumbel noise, which yields the softmax top-1 link. The discussion in Section 2.2 suggests that this is only one possible effective regime: even when the conditional utility model $P_{U|S}$ is unknown, the winner event for large candidate sets is governed by tail behavior, and under approximate max-stability it must fall into one of three tail geometries indexed by ξ (Gumbel $\xi = 0$, Fréchet $\xi > 0$, Weibull $\xi < 0$). From this asymptotic standpoint, diagnosing misspecification means asking how much endpoint geometry is active, and in which choice sets, rather than assigning one geometry globally. This leads to the following empirical question:

Is there systematic endpoint evidence beyond the softmax/Gumbel link, and when does it become important for top-1 prediction?

2.3.1. VISUAL EVIDENCE: ENDPOINT BEHAVIOR IN THE SCORE TAIL

We begin with a direct look at a representative aggregate tail of high negative similarities. Applying the standard Peaks-Over-Threshold (POT) methodology to cosine similarities from a frozen encoder (Figure 3), we fit a Generalized Pareto distribution to exceedances above a high threshold. For this aggregate diagnostic, the fitted shape parameter is $\hat{\xi} = -0.39$, indicating Weibull-type endpoint behavior ($\xi < 0$) in the near-cap tail. This provides evidence for an endpoint component in the observed hard-negative score tail, but it neither identifies the latent utility law nor implies that every finite anchor choice set should use a pure Weibull link. The Gumbel special case ($\xi = 0$) is a poor upper-tail fit: its density (red dashed in Section 2.3.1) assigns positive mass beyond the score ceiling, while the Weibull-GPD (black dashed) correctly drops to zero at the cap. The QQ plot (Section 2.3.1) gives the same message: the Weibull-GPD quantiles lie close to the diagonal, while the Gumbel deviates sharply in the upper tail.

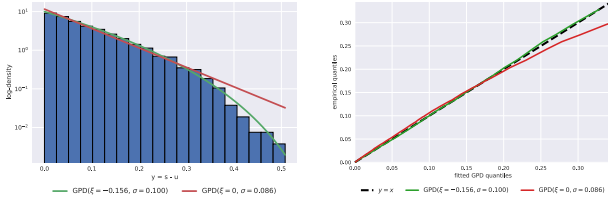


Figure 3. POT analysis of a representative aggregate score tail from a frozen encoder ($\hat{\xi} = -0.39$). **Left:** Exceedance density. Weibull-GPD (black) drops to zero at the cap (green); Gumbel-GPD (red) does not. **Right:** QQ plot. Weibull-GPD (blue) matches the upper tail; Gumbel (red) overestimates upper quantiles.

2.3.2. A LIKELIHOOD-BASED LINK SELECTION TEST

The POT analysis is marginal and tail-level; it does not decide whether every anchor should use a pure endpoint link. We now ask how much this component matters for top-1 prediction relative to the standard translation/Gumbel coordinate. We compare the standard softmax link (translation coordinate) against a nested one-parameter family that interpolates between the translation coordinate and an endpoint (shortfall) coordinate.⁵

Concretely, for each frozen checkpoint, we build an evaluation bank consisting of 32 batches of size $B = 256$, yielding $2B$ augmented views per batch. For each anchor view i in a batch, the candidate set $\mathcal{N}_i$ is the other $2B - 1$ views in the

⁵Heavy tail (Fréchet) behavior is not a natural hypothesis here because cosine similarity scores are bounded above ($s \leq 1$). We therefore focus on translation (Gumbel) versus endpoint (Weibull) regimes in the main experiments. We defer an ablation of the Fréchet case to Section D.6.

same batch, and the observed winner is its paired positive i^+ . We compute cosine similarity scores s_{ik} for $k \in \mathcal{N}_i$ and treat the labels $\{i^+\}$ as top-1 outcomes. This evaluation does not require access to latent utilities.

We split the evaluation bank into held-in batches (used to fit link parameters) and held-out batches (used only for reporting). We compare:

- $\mathcal{H}_0$ (**softmax / Gumbel link**): Under the additive random utility model with i.i.d. Gumbel noise, the top-1 probability is the softmax link

$$p_0(Y_i = j \mid \{s_{ik}\}_{k \in \mathcal{N}_i}) = \frac{\exp(s_{ij}/\tau_0)}{\sum_{k \in \mathcal{N}_i} \exp(s_{ik}/\tau_0)}. \quad (2.7)$$

We fit the temperature $\tau_0 > 0$ by maximizing the held-in top-1 log likelihood.

- $\mathcal{H}_1$ (**blended tail coordinate link**): To capture endpoint geometry, define the shortfall coordinate $g(x) \triangleq -\log(1 - x)$ for $x < 1$ (cosine endpoint $b = 1$), and the nested transform

$$T_\lambda(x) \triangleq (1 - \lambda)x + \lambda g(x), \quad \lambda \in [0, 1]. \quad (2.8)$$

The induced top-1 link is again softmax,

$$p_1(Y_i = j \mid \{s_{ik}\}_{k \in \mathcal{N}_i}) = \frac{\exp(T_\lambda(s_{ij})/\tau_1)}{\sum_{k \in \mathcal{N}_i} \exp(T_\lambda(s_{ik})/\tau_1)}. \quad (2.9)$$

This family is nested since $\lambda = 0$ recovers $\mathcal{H}_0$, while $\lambda = 1$ gives the pure shortfall coordinate. Intermediate values quantify a partial endpoint component rather than an all-or-nothing replacement of softmax. We fit (λ, τ_1) on held-in batches by maximizing the top-1 log likelihood.

Evaluation metric. On held-out batches we report the mean log likelihood gain per anchor,

$$\bar{\Delta} = \mathbb{E}[\log p_1(Y_i = i^+) - \log p_0(Y_i = i^+)],$$

estimated by the held-out sample average (see Section D for details). Since $\bar{\Delta}$ is a mean log ratio, $\exp(\bar{\Delta})$ is the ratio of geometric mean probabilities assigned to the positive under $\mathcal{H}_1$ versus $\mathcal{H}_0$. We assess significance using a one-sided clustered t test over batches, and we also report bootstrap-based confidence intervals (Section D).

Results. Across CIFAR-10, CIFAR-100, and STL-10, and across ResNet-18 and ResNet-50 backbones, the fitted blend weight is consistently nonzero but also far from one ($\hat{\lambda} \approx 0.33$ to 0.50), indicating a systematic endpoint-shortfall component beyond pure translation geometry, rather than a wholesale replacement of softmax (Figure 4). This is

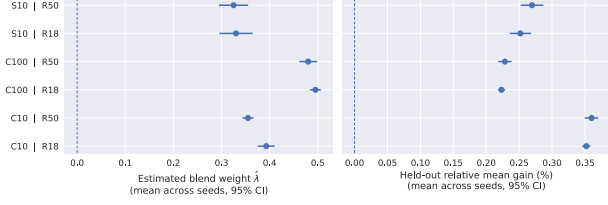


Figure 4. Likelihood-based link selection across datasets and backbones (R18: ResNet18; R50: ResNet50; C10/100: CIFAR10/CIFAR100; S10: STL10). Left: estimated blend weight $\hat{\lambda}$ (mean across seeds, 95% confidence interval). Right: held-out mean log likelihood gain Δ (nats per anchor; mean across seeds, 95% confidence interval). Numeric values are reported in Table 3 in Appendix D; epoch-wise trends (Figure 9) are provided in Section D.

consistent with the FTG view: many anchors contribute near-cap hard negatives, while others remain translation-like or statistically ambiguous. The held-out improvement is substantial ($\bar{\Delta} \approx 0.22$ to 0.36 nats per anchor), corresponding to a geometric mean increase in assigned probability of $\exp(\bar{\Delta}) \approx 1.25$ to 1.43 . The gain is statistically significant under batch-clustered testing and remains stable across training epochs (Section D).

Additional ablations in Section D decouple choice-set hardness from negative-pool size. Including easier negatives dilutes both $\hat{\lambda}$ and $\bar{\Delta}$, while increasing the pool from which a fixed hard set is selected drives $\hat{\lambda}$ upward. This matches the EVT prediction that bounded-endpoint effects become stronger as harder near-cap competitors become available.

Implication for training. The link-selection test is more than a diagnostic of held-out likelihood. A consistently nonzero endpoint weight means that winner prediction contains a systematic endpoint-coordinate component beyond the raw-score softmax family. Under the endpoint random utility model of Section 2.2.2, a pure raw-score softmax link is then a statistically misspecified winner model on the affected score configurations. In the quasi-MLE sense, softmax then targets the closest member of the wrong link family rather than the true top-1 law. This misspecification can lead to unfavorable gradient allocation, because contrastive training places gradient mass according to the fitted winner probabilities. We give the formal White-style misspecification statement, a synthetic noise-floor illustration, and the risk-versus-gradient diagnostic in Section C. The method below uses this evidence constructively: rather than replacing softmax wholesale, WEINCE adapts the correction anchor by anchor.

3. WEINCE: Correcting Softmax at the Extremes

Section 2.3 shows that bounded-similarity top-1 outcomes often contain an endpoint shortfall component, especially for near-cap hard negatives. Because finite minibatches mix endpoint-like, translation-like, and ambiguous anchors, we use an adaptive objective that corrects InfoNCE logits only when anchor-wise tail evidence supports it. The full training step is summarized in Algorithm 1; implementation details are deferred to Section E.

Logit level tail interpolation. Consider a batch of $2N$ ℓ_2 normalized embeddings $\{z_i\}_{i=1}^{2N}$ and cosine scores $s_{ij} \triangleq z_i^\top z_j \in [-1, 1]$. For each anchor i , let i^+ denote its paired positive and let $\mathcal{N}_i = \{1, \dots, 2N\} \setminus \{i\}$ be the candidate set (positive plus negatives).

InfoNCE uses Plackett–Luce (softmax) logits

$$\ell_{ij}^{\text{PL}} \triangleq \frac{s_{ij}}{\tau},$$

while a Weibull shortfall model suggests logits of the form

$$\ell_{ij}^{\text{W}} \triangleq -\beta \log(x_F - s_{ij}) \quad (s_{ij} < 1).$$

with $x_F = 1$ for cosine similarity, according to the tail-index model given in Equation (2.6) and Table 1. We define interpolated logits

$$\ell_{ij} \triangleq (1 - \lambda_i) \ell_{ij}^{\text{PL}} + \lambda_i \ell_{ij}^{\text{W}}, \quad \lambda_i \in [0, 1], \quad (3.1)$$

where λ_i is an anchor-wise interpolation weight and β is estimated online (anchor-wise in our implementation, as described below). This nesting is intentional: $\lambda_i = 0$ recovers vanilla InfoNCE, $\lambda_i = 1$ gives a pure endpoint-shortfall link, and intermediate values represent anchors for which the tail evidence is mixed. The resulting loss is standard cross entropy:

$$\mathcal{L}_{\text{WEINCE}}(\theta) \triangleq -\frac{1}{2N} \sum_{i=1}^{2N} \log \frac{\exp(\ell_{i,i^+})}{\sum_{j \in \mathcal{N}_i} \exp(\ell_{ij})}. \quad (3.2)$$

In practice, λ_i is estimated separately for each anchor rather than fixed globally, so ambiguous anchors remain close to InfoNCE while endpoint-like anchors receive larger shortfall corrections.

Endpoint shortfall for cosine scores. For bounded cosine similarity the endpoint is $x_F = 1$. The Weibull row of Table 1 gives $p_j \propto q(s_{ij})^{-\beta}$. If q is left differentiable at x_F with nonzero endpoint derivative, then locally $q(s) = c(x_F - s) + o(x_F - s)$. We therefore use the corresponding first-order endpoint model $q_0(s) = x_F - s$, which yields $\ell_{ij}^{\text{W}} = -\beta \log(x_F - s_{ij})$. A fuller derivation and the scope of this approximation are given in Section E.

How λ_i and β are chosen. Equation (3.1) requires an anchor-wise mixing weight λ_i and a shortfall slope β . This is where WEINCE differs from a pure Weibit loss: because the active finite-sample tail regime can vary across anchors, λ_i stays close to zero unless the current anchor shows both near-cap and Weibull-like evidence. We estimate these quantities *online* from the current minibatch similarity matrix, using only the negative shortfalls $\delta_{ij} \triangleq 1 - s_{ij}$ for $j \in \mathcal{N}(i)$:

- (i) a *cap-proximity* (hardness) signal $\rho_i = \min_{j \in \mathcal{N}(i)} \delta_{ij}$;
- (ii) a *tail-shape* signal obtained by fitting a Weibull line to the K_{tail} smallest shortfalls and comparing it against a Gumbel proxy fit via an AIC-style score.

We use the fitted Weibull slope $\hat{\beta}_i$ in $\ell_{ij}^W = -\hat{\beta}_i \log(1 - s_{ij})$ and set λ_i as a smooth function increasing in both “near-cap” evidence (small ρ_i) and “Weibull-like” evidence (positive ΔAIC_i). Anchors far from the cap or with Gumbel-like tail fits receive λ_i close to zero, while anchors with strong near-cap and Weibull-like evidence receive larger endpoint corrections. For stability, we clip s_{ij} away from 1 inside $\log(1 - s_{ij})$, and we treat $(\lambda_i, \hat{\beta}_i)$ as `stop_grad` statistics (no backprop through the tail fits). The complete estimation rules for λ_i and $\hat{\beta}_i$ are summarized in Algorithm 1, with additional derivation details in Section E.

Computational overhead. All tail statistics are computed from the batch similarity matrix with no additional forward passes or trainable parameters. In a representative Tiny-ImageNet/ResNet-18 run with batch size 256 on an NVIDIA L40S over 770 steps, the mean step time changes from 253.3 ms for InfoNCE to 255.0 ms for WEINCE, corresponding to a +0.67% end-to-end overhead and a throughput change from 1011 to 1004 samples/s. The loss-stage computation itself increases from 0.65 ms to 2.11 ms because of the per-anchor tail fit, but this stage remains below 1% of total step time.

4. Experiments

We evaluate WEINCE as a drop-in replacement for the standard InfoNCE loss across vision and NLP settings. For vision, we use the official SimCLR implementation (Chen et al., 2020; Spijkervet, 2020), follow the dataset-specific SimCLR settings (augmentations, optimizer, temperature $\tau=0.5$, and training schedule), and modify only the contrastive loss. Most vision runs use ResNet-18 and ResNet-50 encoders with the standard SimCLR projection head; the Tiny-ImageNet rebuttal sweep also includes a ViT-Small encoder. For NLP, we evaluate on unsupervised SimCSE (Gao et al., 2022) with BERT-base-uncased and cosine similarity ($\tau=0.05$), replacing only the logit computation. In both cases, WEINCE introduces no additional trainable param-

Algorithm 1 WEINCE: Full logit interpolation

```

1: Input: Embeddings  $\{z_i\}_{i=1}^{2N}$ , positives  $i^+$ , temp.  $\tau$ 
2: Hyperparams:  $K_{\text{tail}}, \rho_0, m, \kappa_\rho, \kappa_{\text{AIC}}, \varepsilon$ 
3: Calc. similarities  $s_{ij} \leftarrow z_i^\top z_j$  for all  $i \neq j$ 
4: for  $i = 1$  to  $2N$  do
5:    $\mathcal{C}(i) \leftarrow \{1, \dots, 2N\} \setminus \{i\}$ 
6:    $\mathcal{N}(i) \leftarrow \mathcal{C}(i) \setminus \{i^+\}$ 
7:   Calc. shortfalls  $\delta_{ij} \leftarrow 1 - s_{ij}$  for  $j \in \mathcal{N}(i)$ 
8:    $\rho_i \leftarrow \min_{j \in \mathcal{N}(i)} \delta_{ij}$ 
9:   Run TAILFIT on smallest  $K_{\text{tail}}$  shortfalls  $\rightarrow (\hat{\beta}_i, \Delta\text{AIC}_i)$ 
10:   $\lambda_i \leftarrow \sigma(\kappa_\rho \log(\rho_0/\rho_i)) \cdot \sigma(\kappa_{\text{AIC}}(\Delta\text{AIC}_i - m))$ 
11:  for  $j \in \mathcal{C}(i)$  do
12:     $\ell_{ij}^{\text{PL}} \leftarrow s_{ij}/\tau$ 
13:     $\tilde{s}_{ij} \leftarrow \min(s_{ij}, 1 - \varepsilon)$ 
14:     $\ell_{ij}^W \leftarrow -\hat{\beta}_i \log(\varepsilon + 1 - \tilde{s}_{ij})$ 
15:     $\ell_{ij} \leftarrow (1 - \lambda_i)\ell_{ij}^{\text{PL}} + \lambda_i\ell_{ij}^W$ 
16:  end for
17: end for
18: Compute cross entropy loss:
19:  $\mathcal{L} \leftarrow -\frac{1}{2N} \sum_{i=1}^{2N} \log \frac{\exp(\ell_{i,i^+})}{\sum_{j \in \mathcal{C}(i)} \exp(\ell_{ij})}$ 
20: return  $\mathcal{L}$ 

```

Subroutine TAILFIT:

Given K_{tail} smallest shortfalls $\delta_{(1)} \leq \dots \leq \delta_{(K_{\text{tail}})}$:

- Set empirical CDF: $\hat{F}(\delta_{(k)}) = k/(K_{\text{tail}} + 1)$
- Fit Weibull: $\log \hat{F} \approx a + \beta \log \delta \rightarrow \text{get } \hat{\beta}, \text{AIC}_W$
- Fit Gumbel: $\log(-\log \hat{F}) \approx c - \kappa \log \delta \rightarrow \text{get } \text{AIC}_G$
- Return $\Delta\text{AIC} = \text{AIC}_G - \text{AIC}_W$ and $\hat{\beta}$

■ Vanilla InfoNCE

■ WEINCE Additions

ters; the mixing weight λ_i and tail index $\hat{\beta}_i$ are stop-gradient batch statistics.

We evaluate representation quality using two standard frozen-feature protocols: (i) downstream linear evaluation and (ii) k -nearest-neighbor (kNN) evaluation. To avoid tuning on the test set, we create a stratified validation split by holding out 20% of the labeled training set and using the remaining 80% to train the evaluator or build the kNN bank. When multiple hyperparameter configurations are compared, we select the configuration using validation performance only and report the corresponding test-set results.

4.1. Vision benchmarks

Results on CIFAR-10/100 (Krizhevsky & Hinton, 2009), STL-10 (Coates et al., 2011), ImageNet-32 (Deng et al., 2009; Chrabaszcz et al., 2017), and Tiny-ImageNet are summarized by linear accuracy in Table 2. The full kNN recall table is reported in Appendix Table 5. WEINCE consistently improves over InfoNCE across datasets and backbones, with the largest linear gains on CIFAR-100 (+3.27% / +4.82% for ResNet-18/50). The Tiny-ImageNet rebuttal results show the same pattern: ResNet-18 gains mainly in

Table 2. Downstream linear evaluation accuracy (%). We report the mean accuracy $\pm$ 95% confidence intervals comparing the vanilla InfoNCE baseline (InfoNCE) with our proposed method (WEINCE). Encoders are abbreviated as R18 (ResNet18) and R50 (ResNet50). Best results are highlighted in bold.

Dataset	Enc.	Lin. Acc	
		InfoNCE	WEINCE
STL10	R18	76.54 $\pm$ 0.32	77.94 $\pm$ 0.27
	R50	78.44 $\pm$ 0.31	80.02 $\pm$ 0.46
CIFAR10	R18	81.55 $\pm$ 0.40	81.89 $\pm$ 0.30
	R50	82.13 $\pm$ 0.92	83.74 $\pm$ 0.18
CIFAR100	R18	50.01 $\pm$ 0.19	53.28 $\pm$ 0.28
	R50	51.15 $\pm$ 0.43	55.97 $\pm$ 1.21
Imagenet32	R18	24.29 $\pm$ 0.30	25.26 $\pm$ 0.28
Tiny-IN	R18	30.76	30.66

kNN retrieval (R@1 +1.55, R@2 +1.35), while ViT-Small improves both linear accuracy (+3.41%) and kNN recall (R@1 +4.23, R@20 +4.34). Complete evaluation details are provided in Section F.

4.2. Cross-domain generality: sentence embeddings

The theoretical argument for WEINCE applies to any objective that computes a softmax over bounded cosine similarities. To test this beyond vision, we evaluate on unsupervised SimCSE (Gao et al., 2022), which uses InfoNCE with cosine similarity ($\tau=0.05$) over sentence embeddings from BERT-base-uncased. Dropout serves as the augmentation, making the contrastive structure closely parallel to SimCLR. We train on 1M Wikipedia sentences and evaluate on the STS Benchmark (Spearman correlation).

On STS-B, InfoNCE achieves a Spearman score of 71.74 ± 4.23 (reported as Spearman $\times 100$), while WEINCE achieves 76.36 ± 3.17 , a gain of +4.63 points ($p < 0.05$, paired one-sided t -test), with no architecture or training-schedule changes. This suggests that the bounded-endpoint correction is not vision-specific.

5. Conclusion

We interpreted InfoNCE as a top-1 Plackett–Luce likelihood and used extreme value theory to expose the tail-geometry assumption behind its softmax link. For bounded cosine similarities, the hardest negatives often live near a finite endpoint, where Weibull shortfall behavior gives a better local description than a pure translation/Gumbel link. Our diagnostics show that this mismatch affects both held-out top-1 likelihood and gradient allocation. WEINCE addresses the mismatch with a drop-in logit interpolation whose anchor-wise weight and tail index are estimated online from batch shortfalls, adding no trainable parameters and negligible overhead. Across five vision benchmarks and SimCSE sen-

tence embeddings, this endpoint-aware correction consistently improves frozen-feature evaluation.

Impact Statement

This work aims to improve self-supervised representation learning by better modeling hard negatives in contrastive objectives, potentially reducing label requirements and annotation cost. Like other improvements to general-purpose representation learning, it could also strengthen downstream systems with harmful or inequitable uses, including surveillance, profiling, misinformation, or biased decision pipelines. The method does not introduce new data collection or a deployed system, but applications should still be audited for fairness, privacy, robustness, and dataset provenance.

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A. A Permutation Block Extension of FTG

A.1. Scope and setup

The classical Fisher–Tippett–Gnedenko theorem classifies nondegenerate affine limits of maxima of i.i.d. random variables. In the random utility model of the main text, the utility law of candidate i depends on its score s_i , so conditional on a realized score vector the utilities are independent but nonidentically distributed. Arbitrary nonidentical arrays can have extreme value limits outside the generalized extreme value family. We therefore isolate the minimal replication property used by the classical block proof.

Let $X_i \sim F_{s_i}$ be independent, with deterministic score sequence $\mathbf{s} = (s_i)_{i \geq 1}$, and write

$$M_n = \max_{1 \leq i \leq n} X_i, \quad u_n(x) = b_n + a_n x, \quad a_n > 0.$$

For a fixed integer $m \geq 2$, let π be a uniform random permutation of $[mn]$, independent of the utilities, and define the maximum of the first random block of size n by

$$B_{m,n}^\pi = \max_{1 \leq r \leq n} X_{\pi(r)}.$$

Assumption A.1 (Random block representativeness). For the normalizers a_n, b_n and continuous nondegenerate candidate limit G appearing in Theorem A.2, for every fixed integer $m \geq 2$ and every continuity point x of G ,

$$\mathbb{P}\left(\frac{B_{m,n}^\pi - b_n}{a_n} \leq x \mid \mathbf{s}, \pi\right) \xrightarrow[n \rightarrow \infty]{\mathbb{P}_\pi} G(x). \quad (\text{H})$$

Thus, at the normalization for an n candidate problem, a uniformly selected block of n candidates from the first mn candidates has the same limiting maximum law as the original n candidate pool. The proof below only needs the corresponding convergence after averaging over π ; the conditional formulation makes the representativeness interpretation explicit.

Theorem A.2 (Block FTG for nonidentical utilities). *Let $\{X_i\}_{i \geq 1}$ be independent random variables with continuous CDFs F_{s_i} . Suppose there are sequences $a_n > 0$, $b_n \in \mathbb{R}$, and a continuous nondegenerate CDF G such that*

$$\mathbb{P}\left(\frac{M_n - b_n}{a_n} \leq x \mid \mathbf{s}\right) \longrightarrow G(x)$$

at every continuity point x of G . Assume, for every such x with $0 < G(x) < 1$,

$$\max_{1 \leq i \leq n} \bar{F}_{s_i}(b_n + a_n x) \longrightarrow 0, \quad (\text{I})$$

and assume random block representativeness, Assumption A.1, for every fixed integer $m \geq 2$.

Then G is max stable and hence is of generalized extreme value type. Equivalently, there are $\mu \in \mathbb{R}$, $\sigma > 0$, and $\xi \in \mathbb{R}$ such that

$$G(x) = G_\xi \left(\frac{x - \mu}{\sigma} \right),$$

where

$$G_\xi(z) = \begin{cases} \exp\{-(1 + \xi z)^{-1/\xi}\}, & 1 + \xi z > 0, \quad \xi \neq 0, \\ \exp\{-e^{-z}\}, & \xi = 0. \end{cases}$$

Moreover, defining the aggregate survival function

$$\bar{F}_n(t) = \frac{1}{n} \sum_{i=1}^n \bar{F}_{s_i}(t),$$

for any reference point x_0 with $0 < G(x_0) < 1$,

$$\frac{\bar{F}_n(b_n + a_n x)}{\bar{F}_n(b_n + a_n x_0)} \longrightarrow \frac{-\log G(x)}{-\log G(x_0)}$$

locally uniformly on the interior of the support of G .

A.2. Two classical facts

We use the following standard convergence of types result in its correct affine orientation.

Lemma A.3 (Convergence of types). *Suppose*

$$\frac{Y_n - b_n}{a_n} \Rightarrow G, \quad \frac{Y_n - d_n}{c_n} \Rightarrow H,$$

where $a_n, c_n > 0$ and G, H are nondegenerate CDFs. Then there are constants $A > 0$ and $B \in \mathbb{R}$ such that

$$\frac{c_n}{a_n} \longrightarrow A, \quad \frac{d_n - b_n}{a_n} \longrightarrow B,$$

and $H(x) = G(Ax + B)$. at continuity points. This is the classical convergence of types theorem.

We also use the classical characterization that every continuous nondegenerate max stable CDF is of GEV type, and conversely every GEV CDF is max stable; see, for example, [de Haan & Ferreira \(2006\)](#).

Lemma A.4 (Max stability implies GEV). *If for every integer $m \geq 2$ there are $\alpha_m > 0$ and $\beta_m \in \mathbb{R}$ such that*

$$G(x)^m = G(\alpha_m x + \beta_m),$$

then G has the form in Theorem A.2.

A.3. The permutation block argument

Lemma A.5 (Permutation block argument). *Under the hypotheses of Theorem A.2, the limit G is max stable.*

Proof. Fix an integer $m \geq 2$ and a point x with $0 < G(x) < 1$. At the normalization for the mn candidate maximum, set

$$v_n(x) = b_{mn} + a_{mn}x, \\ p_{i,n}(x) = \bar{F}_{s_i}(v_n(x)), \quad 1 \leq i \leq mn.$$

and write

$$S_n(x) = \sum_{i=1}^{mn} p_{i,n}(x), \\ p_{\max,n}(x) = \max_{1 \leq i \leq mn} p_{i,n}(x).$$

By Theorem A.2 along the subsequence mn ,

$$\prod_{i=1}^{mn} (1 - p_{i,n}(x)) \longrightarrow G(x).$$

Condition (I), again along this subsequence, gives $p_{\max,n}(x) \rightarrow 0$. Since, for $0 \leq p \leq p_{\max,n} < 1$,

$$p \leq -\log(1 - p) \leq \frac{p}{1 - p_{\max,n}},$$

we obtain

$$S_n(x) \longrightarrow \psi(x), \quad \psi(x) = -\log G(x).$$

Now let

$$T_{m,n}^\pi(x) = \sum_{r=1}^n p_{\pi(r),n}(x)$$

be the tail mass in the random block. Sampling without replacement gives

$$\mathbb{E}_\pi T_{m,n}^\pi(x) = \frac{1}{m} S_n(x) \longrightarrow \frac{\psi(x)}{m}.$$

The inclusion indicators of a simple random sample have nonpositive pairwise covariances, so

$$\begin{aligned} \text{Var}_\pi(T_{m,n}^\pi(x)) &\leq \frac{1}{m} \sum_{i=1}^{mn} p_{i,n}(x)^2 \\ &\leq \frac{1}{m} p_{\max,n}(x) S_n(x) \longrightarrow 0. \end{aligned}$$

Hence

$$T_{m,n}^\pi(x) \xrightarrow{\mathbb{P}_\pi} \frac{\psi(x)}{m}.$$

Conditional on π , the random block maximum has CDF

$$\begin{aligned} D_{m,n}^\pi(x) &= \mathbb{P}(B_{m,n}^\pi \leq v_n(x) \mid \mathbf{s}, \pi) \\ &= \prod_{r=1}^n (1 - p_{\pi(r),n}(x)). \end{aligned}$$

The same logarithmic bounds give

$$T_{m,n}^\pi(x) \leq -\log D_{m,n}^\pi(x) \leq \frac{T_{m,n}^\pi(x)}{1 - p_{\max,n}(x)}.$$

Together with the preceding convergence, this implies

$$D_{m,n}^\pi(x) \xrightarrow{\mathbb{P}_\pi} \exp\left\{-\frac{\psi(x)}{m}\right\} = G(x)^{1/m}.$$

Because $0 \leq D_{m,n}^\pi(x) \leq 1$, averaging over π yields

$$\mathbb{P}\left(\frac{B_{m,n}^\pi - b_{mn}}{a_{mn}} \leq x \mid \mathbf{s}\right) \rightarrow G(x)^{1/m}.$$

Assumption A.1, followed by the same bounded convergence argument, gives

$$\mathbb{P}\left(\frac{B_{m,n}^\pi - b_n}{a_n} \leq x \mid \mathbf{s}\right) \rightarrow G(x).$$

Thus the same sequence $B_{m,n}^\pi$ has two affine normalizations with nondegenerate limits G and $G^{1/m}$. By Lemma A.3, there are $A_m > 0$ and $B_m \in \mathbb{R}$ such that

$$G(x)^{1/m} = G(A_m x + B_m).$$

Writing $y = A_m x + B_m$ gives

$$G(y)^m = G\left(\frac{y - B_m}{A_m}\right).$$

Therefore G is max stable, with $\alpha_m = A_m^{-1}$ and $\beta_m = -B_m/A_m$. $\square$

Proof of Theorem A.2. By Lemma A.5, G is max stable. The GEV representation in Theorem A.2 follows from Lemma A.4.

For the aggregate tail statement, define

$$\begin{aligned} p_{i,n}(x) &= \bar{F}_{s_i}(b_n + a_n x), \\ \Lambda_n(x) &= \sum_{i=1}^n p_{i,n}(x), \\ p_{\max,n}(x) &= \max_{1 \leq i \leq n} p_{i,n}(x). \end{aligned}$$

The product formula and condition (I) give, locally uniformly on compact subsets of the interior support,

$$\Lambda_n(x) + \log \mathbb{P}(M_n \leq b_n + a_n x \mid \mathbf{s}) \rightarrow 0.$$

Indeed, on a compact interval its left endpoint controls the maximum individual tail probability, and

$$\begin{aligned} 0 &\leq -\sum_i \log(1 - p_{i,n}(x)) - \sum_i p_{i,n}(x) \\ &\leq \frac{p_{\max,n}(x)}{1 - p_{\max,n}(x)} \sum_i p_{i,n}(x). \end{aligned}$$

Consequently,

$$\Lambda_n(x) \rightarrow -\log G(x)$$

locally uniformly. Since $n\bar{F}_n(b_n + a_n x) = \Lambda_n(x)$, division by the corresponding limit at x_0 proves Theorem A.2. $\square$

A.4. Canonical tail ratio form

The ratio in Theorem A.2 is the correct statement for arbitrary normalizers. To recover the canonical expression used in the main text, write

$$G(x) = G_\xi\left(\frac{x - \mu}{\sigma}\right), \quad G_\xi(z) = \exp\{-\nu_\xi(z)\},$$

where

$$\nu_\xi(z) = \begin{cases} \exp(-z), & \xi = 0, \\ (1 + \xi z)^{-1/\xi}, & \xi \neq 0, \end{cases}$$

on the corresponding support. Set

$$\tilde{b}_n = b_n + \mu a_n, \quad \tilde{a}_n = \sigma a_n.$$

Then

$$\frac{M_n - \tilde{b}_n}{\tilde{a}_n} \Rightarrow G_\xi,$$

and, because $-\log G_\xi(0) = 1$,

$$\frac{\bar{F}_n(\tilde{b}_n + \tilde{a}_n z)}{\bar{F}_n(\tilde{b}_n)} \rightarrow \nu_\xi(z).$$

No density convergence is inferred by differentiating Theorem A.2; the density profile required for winner probabilities is imposed separately in Assumption B.1.

A.5. A conditional FTG alternative under random score indexing

The block theorem treats a deterministic heterogeneous score array. The population InfoNCE model in Definition 2.1 also permits a shorter conditional argument because negatives are sampled i.i.d.

Proposition A.6 (Conditional FTG under i.i.d. score indexing). *Fix an anchor and let $S_1, S_2, \dots \stackrel{\text{i.i.d.}}{\sim} Q$. Conditional on the scores, let*

$$X_i \mid S_i = s \sim F_s$$

independently across i . Suppose, for every continuity point x of a deterministic continuous nondegenerate CDF G ,

$$\prod_{i=1}^n F_{S_i}(b_n + a_n x) \xrightarrow{\mathbb{P}_S} G(x).$$

Then G is of GEV type.

Proof. Define the score averaged utility CDF

$$F_Q(t) = \mathbb{E}_{S \sim Q}[F_S(t)].$$

The conditional maximum CDF in Proposition A.6 is bounded by one, so convergence in probability to the deterministic value $G(x)$ implies convergence of expectations. Independence of the scores gives

$$\begin{aligned} F_Q(b_n + a_n x)^n &= \prod_{i=1}^n \mathbb{E}[F_{S_i}(b_n + a_n x)] \\ &= \mathbb{E}\left[\prod_{i=1}^n F_{S_i}(b_n + a_n x)\right] \longrightarrow G(x). \end{aligned}$$

This is the classical i.i.d. FTG setting for the fixed mixture CDF F_Q , so G is GEV. $\square$

Remark A.7 (Relation between the two arguments). Under i.i.d. score indexing, a uniformly selected block of n coordinates from $(S_1, \dots, S_{mn})$ has the same marginal law as $(S_1, \dots, S_n)$. Thus exchangeability gives the marginal content of Assumption A.1; its conditional-in- π formulation additionally asks the realized reservoir to self-average. The conditional result above gives an alternative route that does not require verifying this property for a deterministic score sequence.

B. Asymptotics of Top-1 Probability

We start by giving the assumptions needed for Theorem 2.4. Then, we restate the result, and give its proof.

Assumption B.1. Let $b_n \in \mathbb{R}$ and $a_n > 0$ be sequences as defined in Theorem 2.4. Define

$$w_{i,n} \triangleq \bar{F}_{s_{i,n}}(b_n) = \mathbb{P}(X_{i,n} > b_n \mid \mathbf{s}_n), \quad \Lambda_n \triangleq \sum_{i \in \mathcal{C}_n} w_{i,n}.$$

There exist an interval $D = (l, r)$ (with $-\infty \leq l < r \leq \infty$), functions $\bar{\nu} : D \rightarrow (0, \infty)$ and $v : D \rightarrow (0, \infty)$, and an envelope $\Phi \in L^1(D)$ such that:

(A1) **Rare exceedances and bounded total mass:**

$$\max_{i \in \mathcal{C}_n} w_{i,n} \rightarrow 0, \quad 0 < \liminf_n \Lambda_n \leq \limsup_n \Lambda_n < \infty.$$

(A2) **Common overshoot shape on scale a_n :** for every compact $K \subset D$,

$$\begin{aligned} \sup_{i \in \mathcal{C}_n} \sup_{x \in K} \left| \frac{\bar{F}_{s_{i,n}}(b_n + a_n x)}{\bar{F}_{s_{i,n}}(b_n)} - \bar{\nu}(x) \right| &\rightarrow 0, \\ \sup_{i \in \mathcal{C}_n} \sup_{x \in K} \left| \frac{a_n f_{s_{i,n}}(b_n + a_n x)}{\bar{F}_{s_{i,n}}(b_n)} - v(x) \right| &\rightarrow 0. \end{aligned}$$

(A3) $\bar{\nu}$ is absolutely continuous and $v = -\bar{\nu}'$: $\bar{\nu}$ is absolutely continuous on D and

$$\bar{\nu}(x) = \int_{t=x}^r v(t) dt, \quad x \in D,$$

with $\bar{\nu}(x) \rightarrow \infty$ as $x \downarrow l$ and $\bar{\nu}(x) \rightarrow 0$ as $x \uparrow r$.

(A4) **Envelope bound:** for all large n , all $i \in \mathcal{C}_n$ and all $x \in D$,

$$0 \leq \frac{a_n f_{s_{i,n}}(b_n + a_n x)}{\bar{F}_{s_{i,n}}(b_n)} \prod_{j \in \mathcal{C}_n \setminus \{i\}} F_{s_{j,n}}(b_n + a_n x) \leq \Phi(x).$$

(A5) **Localization to the extreme window:** uniformly in $i \in \mathcal{C}_n$,

$$\int_{\mathbb{R} \setminus D} \frac{a_n f_{s_{i,n}}(b_n + a_n x)}{\bar{F}_{s_{i,n}}(b_n)} \prod_{j \in \mathcal{C}_n \setminus \{i\}} F_{s_{j,n}}(b_n + a_n x) dx \rightarrow 0.$$

Now, we restate Theorem 2.4, and prove it.

Theorem 2.4 (Tail mass proportionality). *Let $b_n \in \mathbb{R}$ and $a_n > 0$ be as in Theorem A.2. Then under Assumption B.1, uniformly in $1 \leq i \leq n$,*

$$p_i \triangleq \mathbb{P}(I_n = i \mid \mathbf{s}_n) = \frac{\bar{F}_{s_i}(b_n)}{\sum_{r=1}^n \bar{F}_{s_r}(b_n)} (1 + o(1)).$$

Proof. Let us denote the tail mass with

$$w_{i,n} \triangleq \mathbb{P}(X_{i,n} > b_n \mid \mathbf{s}_n) = \bar{F}_{s_{i,n}}(b_n).$$

Conditioning on $X_{i,n}$ gives

$$\mathbb{P}(I_n = i \mid \mathbf{s}_n) = \int_{-\infty}^{\infty} f_{s_{i,n}}(u) \prod_{j \in \mathcal{C}_n \setminus \{i\}} F_{s_{j,n}}(u) du.$$

Change variables $u = b_n + a_n x$, split the integral over D and its complement, and factor out $w_{i,n} = \bar{F}_{s_{i,n}}(b_n)$. By (A5),

$$\mathbb{P}(I_n = i \mid \mathbf{s}_n) = w_{i,n} \left(\int_D G_{i,n}(x) dx + o(1) \right),$$

where

$$G_{i,n}(x) \triangleq \frac{a_n f_{s_{i,n}}(b_n + a_n x)}{\bar{F}_{s_{i,n}}(b_n)} \prod_{j \in \mathcal{C}_n \setminus \{i\}} F_{s_{j,n}}(b_n + a_n x).$$

Fix $x \in D$ and write $p_{j,n}(x) := \bar{F}_{s_{j,n}}(b_n + a_n x)$. By (A2),

$$p_{j,n}(x) = w_{j,n}(\bar{\nu}(x) + o(1))$$

uniformly in j . Since x is fixed and $\bar{\nu}$ is finite on compacts inside D , (A1) implies $\max_j p_{j,n}(x) \rightarrow 0$ and

$$\sum_{j \in \mathcal{C}_n} p_{j,n}(x) = \bar{\nu}(x) \Lambda_n + o(1).$$

where we denote:

$$\Lambda_n \triangleq \sum_{j \in \mathcal{C}_n} w_{j,n}$$

as introduced in Assumption B.1.

For all large n , $\max_j p_{j,n}(x) \leq 1/2$, and using $\log(1-y) = -y + O(y^2)$ uniformly on $y \in [0, 1/2]$,

$$\begin{aligned} \sum_{j \neq i} \log F_{s_{j,n}}(b_n + a_n x) &= \sum_{j \neq i} \log(1 - p_{j,n}(x)) \\ &= - \sum_{j \neq i} p_{j,n}(x) + O\left(\sum_{j \neq i} p_{j,n}(x)^2\right), \end{aligned}$$

and

$$\sum_{j \neq i} p_{j,n}(x)^2 \leq \left(\max_j p_{j,n}(x)\right) \sum_{j \neq i} p_{j,n}(x) \rightarrow 0.$$

Therefore,

$$\prod_{j \neq i} F_{s_{j,n}}(b_n + a_n x) = \exp(-\bar{\nu}(x) \Lambda_n) (1 + o(1)).$$

Also by (A2),

$$\frac{a_n f_{s_{i,n}}(b_n + a_n x)}{\bar{F}_{s_{i,n}}(b_n)} \rightarrow v(x).$$

Hence for each fixed $x \in D$,

$$\sup_{i \in \mathcal{C}_n} \left| G_{i,n}(x) - v(x) e^{-\Lambda_n \bar{\nu}(x)} \right| \rightarrow 0.$$

By (A1), there exist $0 < \underline{\Lambda} < \bar{\Lambda} < \infty$ such that $\underline{\Lambda} \leq \Lambda_n \leq \bar{\Lambda}$ for all large n , hence

$$0 \leq v(x) e^{-\Lambda_n \bar{\nu}(x)} \leq v(x) e^{-\underline{\Lambda} \bar{\nu}(x)}.$$

Moreover, by (A4), $0 \leq G_{i,n}(x) \leq \Phi(x)$ on D for all large n . Thus,

$$\left| G_{i,n}(x) - v(x) e^{-\Lambda_n \bar{\nu}(x)} \right| \leq \Phi(x) + v(x) e^{-\underline{\Lambda} \bar{\nu}(x)}.$$

Since $\Phi \in L^1(D)$ and, by (A3),

$$\int_D v(x) e^{-\underline{\Lambda} \bar{\nu}(x)} dx = \frac{1}{\underline{\Lambda}} \left[e^{-\underline{\Lambda} \bar{\nu}(r^-)} - e^{-\underline{\Lambda} \bar{\nu}(l^+)} \right] = \frac{1}{\underline{\Lambda}},$$

dominated convergence yields

$$\sup_{i \in \mathcal{C}_n} \left| \int_D G_{i,n}(x) dx - \int_D v(x) e^{-\Lambda_n \bar{\nu}(x)} dx \right| \rightarrow 0.$$

Therefore, uniformly in i ,

$$\mathbb{P}(I_n = i \mid \mathbf{s}_n) = w_{i,n} \left(\int_D v(x) e^{-\Lambda_n \bar{\nu}(x)} dx + o(1) \right).$$

By (A3), $\bar{\nu}'(x) = -v(x)$ a.e. on D , so

$$\frac{d}{dx} e^{-\Lambda_n \bar{\nu}(x)} = \Lambda_n v(x) e^{-\Lambda_n \bar{\nu}(x)}.$$

Integrating over D and using the boundary limits gives

$$\Lambda_n \int_D v(x) e^{-\Lambda_n \bar{\nu}(x)} dx = e^{-\Lambda_n \bar{\nu}(r^-)} - e^{-\Lambda_n \bar{\nu}(l^+)} = 1 - 0 = 1.$$

Hence $\int_D v(x) e^{-\Lambda_n \bar{\nu}(x)} dx = 1/\Lambda_n$, and thus $\mathbb{P}(I_n = i \mid \mathbf{s}_n) = w_{i,n}/\Lambda_n (1 + o(1))$ uniformly in i . $\square$

B.1. Computing tail masses from tail ratios under geometry-matched indexing

The previous proposition provides *tail ratios* in the extreme-value window. To apply Theorem 2.4 we also need a common-factor representation for the *tail masses* $w_{i,n} = \bar{F}_{s_{i,n}}(b_n)$ across the index i . The following lemma records the corresponding calculations under the three canonical geometry-matched index models (Table 1 in the main text).

Theorem B.2 (Tail masses under geometry-matched index models). *Let $\{s_{i,n}\}_{i \in \mathcal{C}_n}$ be deterministic scores and let $w_{i,n} := \bar{F}_{s_{i,n}}(b_n)$.*

(a) **Translation (Gumbel-type indexing).** *Assume $F_s(x) = F_0(x - s)$ for a continuous baseline F_0 . Assume there exist $b_n \in \mathbb{R}$, $a_n > 0$ and a function $\bar{\nu}$ such that for every compact $K \subset D$,*

$$\sup_{x \in K} \left| \frac{\bar{F}_0(b_n + a_n x)}{\bar{F}_0(b_n)} - \bar{\nu}(x) \right| \rightarrow 0.$$

If $t_{i,n} := s_{i,n}/a_n$ satisfy $\sup_{i \in \mathcal{C}_n} |t_{i,n}| \leq T$ for some $T < \infty$ with $[-T, T] \subset D$, then uniformly in $i \in \mathcal{C}_n$,

$$w_{i,n} = \bar{F}_0(b_n) \bar{\nu}(-t_{i,n}) (1 + o(1)).$$

In particular, if $\bar{\nu}(x) = e^{-x}$ (Gumbel), then $w_{i,n} = \bar{F}_0(b_n) \exp(s_{i,n}/a_n) (1 + o(1))$.

(b) **Scaling (Fréchet-type indexing).** *Assume $X_s = \rho(s) Z$ where Z is a nonnegative baseline random variable with tail $\bar{F}_Z$ and $\rho(s) > 0$. Assume there exist $b_n \rightarrow \infty$ and $\alpha > 0$ such that for every compact $K \subset (0, \infty)$,*

$$\sup_{y \in K} \left| \frac{\bar{F}_Z(b_n y)}{\bar{F}_Z(b_n)} - y^{-\alpha} \right| \rightarrow 0.$$

If $\rho(s_{i,n}) \in [\rho_{\min}, \rho_{\max}]$ for some $0 < \rho_{\min} \leq \rho_{\max} < \infty$, then uniformly in $i \in \mathcal{C}_n$,

$$w_{i,n} = \bar{F}_Z(b_n) \rho(s_{i,n})^\alpha (1 + o(1)).$$

(c) **Endpoint shortfall (Weibull-type indexing).** *Assume there is a common right endpoint $b \in \mathbb{R}$ and*

$$b - X_s = q(s) Z,$$

where $q(s) > 0$ and $Z \geq 0$ is a baseline random variable with CDF H . Assume H is continuous near zero and, for some $\beta > 0$ and $\lambda > 0$,

$$\frac{H(z)}{\lambda z^\beta} \rightarrow 1 \quad \text{as } z \downarrow 0.$$

Let $t_n \downarrow 0$ satisfy

$$\max_{i \in \mathcal{C}_n} \frac{t_n}{q(s_{i,n})} \rightarrow 0,$$

and set $b_n := b - t_n$. Then uniformly in $i \in \mathcal{C}_n$,

$$w_{i,n} = \mathbb{P}(X_{s_{i,n}} > b - t_n) = \lambda t_n^\beta q(s_{i,n})^{-\beta} (1 + o(1)).$$

Proof. (a) By the translation model,

$$w_{i,n} = \bar{F}_{s_{i,n}}(b_n) = \bar{F}_0(b_n - s_{i,n}) = \bar{F}_0(b_n + a_n(-t_{i,n})).$$

Since $-t_{i,n} \in [-T, T] \subset D$ uniformly, the assumed tail ratio gives

$$\frac{w_{i,n}}{\bar{F}_0(b_n)} = \frac{\bar{F}_0(b_n + a_n(-t_{i,n}))}{\bar{F}_0(b_n)} = \bar{\nu}(-t_{i,n}) + o(1)$$

uniformly in i .

(b) By $X_s = \rho(s)Z$,

$$w_{i,n} = \mathbb{P}(\rho(s_{i,n})Z > b_n) = \bar{F}_Z\left(\frac{b_n}{\rho(s_{i,n})}\right).$$

Set $y_{i,n} := 1/\rho(s_{i,n}) \in [1/\rho_{\max}, 1/\rho_{\min}]$, a compact subset of $(0, \infty)$. Then

$$\frac{w_{i,n}}{\bar{F}_Z(b_n)} = \frac{\bar{F}_Z(b_n y_{i,n})}{\bar{F}_Z(b_n)} = y_{i,n}^{-\alpha} + o(1) = \rho(s_{i,n})^\alpha + o(1),$$

uniformly in i .

(c) By $b - X_s = q(s)Z$,

$$\begin{aligned} w_{i,n} &= \mathbb{P}(X_{s_{i,n}} > b - t_n) = \mathbb{P}(b - X_{s_{i,n}} < t_n) \\ &= \mathbb{P}\left(Z < \frac{t_n}{q(s_{i,n})}\right) = H\left(\frac{t_n}{q(s_{i,n})}\right) \end{aligned}$$

Since $\max_i t_n/q(s_{i,n}) \rightarrow 0$, the lower-tail expansion of H holds uniformly over the active candidates and yields

$$w_{i,n} = \lambda \left(\frac{t_n}{q(s_{i,n})}\right)^\beta (1 + o(1)) = \lambda t_n^\beta q(s_{i,n})^{-\beta} (1 + o(1)),$$

uniformly in i . $\square$

C. Geometry mismatch and misspecification for top-1 links

C.1. Quasi maximum likelihood under link misspecification

In the main text we compare different parametric top 1 links $p_\theta(\cdot | \mathbf{s})$ for winner observations. It is helpful to separate two cases.

Correct specification. If the data are generated from the assumed link family, meaning there exists a parameter θ_* such that $p_*(\cdot | \mathbf{s}) = p_{\theta_*}(\cdot | \mathbf{s})$ almost surely, then maximum likelihood targets θ_* and, under standard regularity conditions, the estimator is asymptotically efficient.

Misspecification and quasi maximum likelihood. If the true conditional winner law $p_*(\cdot | \mathbf{s})$ is not contained in the assumed family, then the maximum likelihood estimator is still well defined, but it no longer targets a true parameter. Instead, it converges to a *pseudo true* parameter

$$\theta^* \in \arg \max_{\theta} \mathbb{E}[\log p_\theta(Y | \mathbf{s})],$$

which is the best approximation to p_* within the chosen link family (equivalently, it minimizes a KL divergence in expectation). In this case the estimator is called a quasi maximum likelihood estimator, and its large sample covariance is given by a robust sandwich expression. We refer to (White, 1982; Gouriou et al., 1984; van der Vaart, 1998) for general statements and conditions. The formal statement is Theorem C.1; we include the proof sketch here for completeness.

Theorem C.1 (Quasi maximum likelihood under misspecification, after White). *Let $Z_j = (Y_j, \mathbf{s}_j)$ be i.i.d. from P_* , and let $p_\theta(\cdot | \mathbf{s})$ be a possibly misspecified conditional top-1 link. If $\hat{\theta}_m \in \arg \max_{\theta \in \Theta} \sum_{j=1}^m \log p_\theta(Y_j | \mathbf{s}_j)$, then under standard regularity conditions $\hat{\theta}_m \rightarrow \theta^*$, where*

$$\theta^* \in \arg \max_{\theta \in \Theta} \mathbb{E}_{P_*} \log p_\theta(Y | \mathbf{s}),$$

and

$$\sqrt{m}(\hat{\theta}_m - \theta^*) \Rightarrow \mathcal{N}(0, H^{-1} J H^{-1}).$$

Here, with $\ell_\theta(Z) = \log p_\theta(Y | \mathbf{s})$, $H = -\mathbb{E} \nabla_\theta^2 \ell_{\theta^*}(Z)$ and $J = \mathbb{E}[\nabla_\theta \ell_{\theta^*}(Z) \nabla_\theta \ell_{\theta^*}(Z)^\top]$. Correct specification gives $J = H$; under misspecification the estimator targets the pseudo-true KL projection.

Proof sketch of Theorem C.1. This is a standard result for M estimators. With $\ell_\theta(Z) = \log p_\theta(Y | \mathbf{s})$, the score equation $\sum_{j=1}^m \nabla_\theta \ell_{\hat{\theta}_m}(Z_j) = 0$ is expanded around θ^* via a Taylor approximation. A law of large numbers controls the Hessian term and a multivariate central limit theorem controls the score term, yielding the stated asymptotic normality

with sandwich covariance. See (White, 1982; van der Vaart, 1998; Gourieroux et al., 1984). $\square$

Finite-sample illustration. The theorem says that a misspecified likelihood can become statistically stable around the wrong object: the pseudo-true KL projection. Figure 5 illustrates this phenomenon in the same top-1 language used in the paper. Data are generated from an endpoint-shortfall law, and the target is the true top-vs-reference log-odds. The correctly specified Weibit estimator has the expected $\mathcal{O}(1/n)$ error decay, while the misspecified one-parameter softmax estimator converges to its pseudo-true limit and retains a nonzero asymptotic error floor. Thus Figure 5 should be read as a concrete demonstration of Theorem C.1, specialized to the link misspecification that arises when endpoint geometry is forced into a softmax coordinate.

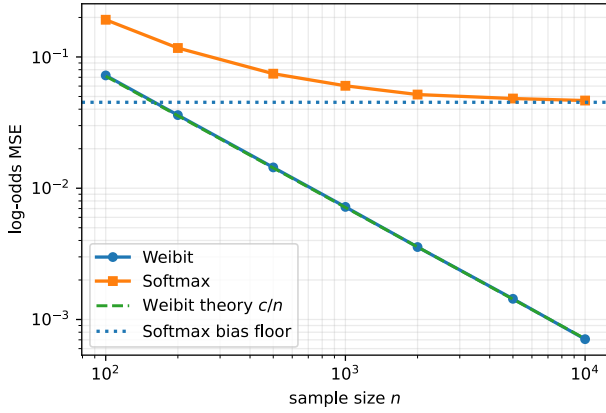


Figure 5. Noise floor from link misspecification in a synthetic top-1 model. Data are generated from an endpoint-shortfall (Weibit) law and the target is the true top-vs-reference log-odds. The correctly specified Weibit estimator has error decaying as $\Theta(1/n)$, while the misspecified one-parameter softmax converges to a pseudo-true parameter and retains a nonzero asymptotic error floor. This illustrates the practical consequence of Theorem C.1.

From estimation to training. The noise-floor example is an estimation-level consequence: the wrong link converges, but to the wrong target. In contrastive learning, the same top-1 link also determines how each batch allocates optimization pressure. The next subsection translates the same misspecification into a gradient-level diagnostic.

C.2. Gradient misallocation under link misspecification

In our setting, $p_\theta(Y | s)$ is the top-1 link used by the contrastive loss. Thus a systematic held-out gain for the endpoint-blended link means that the softmax fit should be viewed as a pseudo-true projection rather than the data-generating winner law. For training, this projection has an operational consequence: it determines where gradient mass is placed. The InfoNCE gradient with respect to negative j

is $\partial\mathcal{L}/\partial s_j = p_j^{\text{SM}}/\tau$, where $1/\tau$ is constant across negatives. The normalized softmax gradient is therefore exactly the model’s predicted win-probability distribution $\{p_j^{\text{SM}}\}$. For anchors whose top-1 outcomes are well described by the softmax/Gumbel link, this distribution should align with the empirical failure distribution. Systematic divergence is evidence that a pure softmax link is misspecified for a nontrivial part of the batch.

To measure this, we freeze an encoder and, for each anchor, repeatedly sample random subsets of K negatives from a large bank of $N \approx 1536$. Whenever a negative beats the positive (a “failure”), we record the cosine similarity of the culprit, giving the **risk distribution** (black curve in Figure 6). On the same subsets, we compute the gradient magnitude of InfoNCE and of a pure endpoint-shortfall (Weibit) loss with respect to every negative and bin by score, giving the **gradient distributions** (red = softmax, blue = Weibit). The blue curve is a diagnostic reference for the endpoint regime; the training objective in Section 3 decides anchor by anchor how much of this component to use.

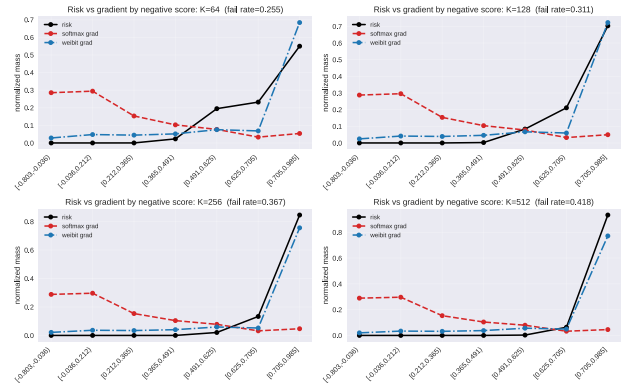


Figure 6. Risk vs. gradient distributions, binned by negative cosine similarity. Black: where failures come from. Red: where InfoNCE allocates gradient. Blue: where a pure endpoint-shortfall (Weibit) diagnostic allocates gradient. Each panel is a different choice-set size K .

Figure 6 reveals two facts. First, InfoNCE misallocates gradient mass: the risk (black) concentrates in the highest score bins, while the softmax gradient (red) spreads substantial mass across low and medium score bins where the empirical failure risk is nearly zero. This happens because the exponential function assigns non-trivial probability to every negative; with hundreds of easy negatives each getting a small share, their combined gradient mass can drown out the few genuinely dangerous hard ones. Second, the pure Weibit diagnostic (blue) is much closer to the empirical risk because it concentrates gradient mass near the finite endpoint. It is not identical to the black curve, however: depending on K , the empirical risk may retain mass in the second-highest bin or be even more concentrated in the fi-

nal bin than the Weibit gradient. This residual mismatch is expected from the theory: FTG describes the extreme-tail domain of attraction, whereas a finite minibatch contains a mixture of anchors, some with near-cap competitors and some whose effective top-1 behavior is still closer to the translation coordinate. Thus the figure is evidence for a substantial endpoint component, not evidence that pure Weibit should replace softmax for every anchor. This motivates the corrected objective in Section 3, which preserves the softmax link when endpoint evidence is weak and redirects gradient mass toward the near-cap regime when the batch tail is Weibull-like.

C.3. Coordinate mismatch implies link misspecification

We next record the elementary coordinate-mismatch facts that justify the link-misspecification interpretation used above. Changing the score coordinate inside a softmax generally changes the induced top 1 link in a way that cannot be fixed by re tuning the temperature. Thus, when the extreme tail is naturally described by an endpoint-shortfall coordinate, forcing it into the translation coordinate creates a genuine top-1 link mismatch.

We use the following shorthand for a softmax top-1 link in a score coordinate T :

$$p_i^{(T,\tau)}(s) \triangleq \frac{\exp(T(s_i)/\tau)}{\sum_{j=0}^K \exp(T(s_j)/\tau)}, \quad i \in \{0, \dots, K\},$$

where $s = (s_0, \dots, s_K)$ and $\tau > 0$.

Example 1 (Translation vs endpoint coordinate cannot be absorbed by temperature). Let $T_{\text{tr}}(s) = s$ (translation coordinate) and $T_{\text{ep}}(s) = -\log(1 - s)$ for $s < 1$ (endpoint shortfall coordinate). Consider three candidates with scores $(s_0, s_1, s_2) = (0, 0.5, 0.9)$. For any coordinate T and any temperature τ , the ratio of pairwise log-odds

$$R_T(s) \triangleq \frac{\log(p_1^{(T,\tau)}(s)/p_0^{(T,\tau)}(s))}{\log(p_2^{(T,\tau)}(s)/p_0^{(T,\tau)}(s))} = \frac{T(s_1) - T(s_0)}{T(s_2) - T(s_0)}$$

is independent of τ . A direct computation gives

$$R_{T_{\text{tr}}}(s) = \frac{0.5 - 0}{0.9 - 0} = \frac{5}{9},$$

$$R_{T_{\text{ep}}}(s) = \frac{-\log(1 - 0.5)}{-\log(1 - 0.9)} = \frac{\log 2}{\log 10}$$

Since $\frac{5}{9} \neq \frac{\log 2}{\log 10}$, there is no choice of temperatures $\tau_{\text{tr}}, \tau_{\text{ep}} > 0$ such that $p^{(T_{\text{tr}}, \tau_{\text{tr}})}(s) = p^{(T_{\text{ep}}, \tau_{\text{ep}})}(s)$. In particular, a softmax link in the translation coordinate cannot reproduce a softmax link in the endpoint coordinate on score triples of this form.

Lemma C.2 (Softmax coordinate equivalence forces an affine transform). *Let $I \subset \mathbb{R}$ and let $T, \tilde{T} : I \rightarrow \mathbb{R}$ be*

functions. Assume there exist temperatures $\tau, \tilde{\tau} > 0$ such that for every $K \geq 1$ and every score vector $s \in I^{K+1}$,

$$p^{(T,\tau)}(s) = p^{(\tilde{T},\tilde{\tau})}(s).$$

Then there exist constants $a > 0$ and $b \in \mathbb{R}$ such that

$$T(s) = a\tilde{T}(s) + b, \quad \forall s \in I,$$

with $a = \tau/\tilde{\tau}$.

Proof. It suffices to consider $K = 1$ (two candidates). Equality of the two links implies that for all $s_0, s_1 \in I$,

$$\frac{p_0^{(T,\tau)}(s_0, s_1)}{p_1^{(T,\tau)}(s_0, s_1)} = \frac{p_0^{(\tilde{T},\tilde{\tau})}(s_0, s_1)}{p_1^{(\tilde{T},\tilde{\tau})}(s_0, s_1)}.$$

By the softmax form,

$$\log\left(\frac{p_0^{(T,\tau)}(s_0, s_1)}{p_1^{(T,\tau)}(s_0, s_1)}\right) = \frac{T(s_0) - T(s_1)}{\tau}, \quad (\text{C.1})$$

$$\log\left(\frac{p_0^{(\tilde{T},\tilde{\tau})}(s_0, s_1)}{p_1^{(\tilde{T},\tilde{\tau})}(s_0, s_1)}\right) = \frac{\tilde{T}(s_0) - \tilde{T}(s_1)}{\tilde{\tau}} \quad (\text{C.2})$$

Therefore, for all $s_0, s_1 \in I$,

$$T(s_0) - T(s_1) = \frac{\tau}{\tilde{\tau}}(\tilde{T}(s_0) - \tilde{T}(s_1)).$$

Fix any $s_{\text{ref}} \in I$ and set $s_1 = s_{\text{ref}}$. Writing $a \triangleq \tau/\tilde{\tau}$ and $b \triangleq T(s_{\text{ref}}) - a\tilde{T}(s_{\text{ref}})$ yields

$$T(s) = a\tilde{T}(s) + b, \quad \forall s \in I.$$

□

Corollary C.3 (Mismatch in tail coordinate implies misspecification). *Let $\tilde{T}(s) = s$ and let T be any non-affine transform on I . Then there is no temperature $\tau > 0$ such that the link $p^{(T,\tau)}$ can be represented as $p^{(\tilde{T},\tau)}$ uniformly over all candidate sets and score vectors.*

D. Likelihood Ratio Test Details

This appendix describes the likelihood-based link selection procedure used in Section 2.3. Table 3 shows the actual numbers in Figure 4. As a complementary analysis to Figure 4, we display the $\hat{\lambda}$ and Δ across epochs in Figure 9.

D.1. Hard-negative and pool-size ablations

The main text reports the aggregate link-selection result. Here we decouple two effects: the hardness of the negatives included in the choice set and the size of the pool from which those hard negatives are drawn. When the total pool is fixed at $K = 512$, adding progressively easier negatives reduces both the fitted endpoint weight and the held-out likelihood gain (Figure 7). When the hard set is fixed at $K_{\text{neg}} = 64$ and the pool size grows, the selected hard negatives become closer to the endpoint and the fitted endpoint weight increases (Figure 8).

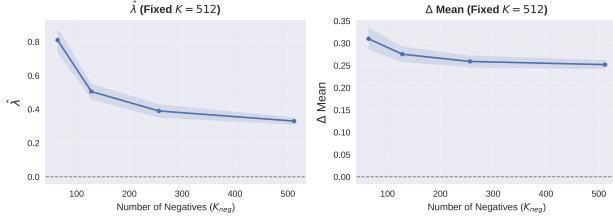


Figure 7. Decoupling the choice set size from the negative pool (fixed $K = 512$). For each anchor, we form a negative pool of size $K=512$, then construct the choice set by hard selecting K_{neg} negatives (largest similarity) and adding the positive. Left: fitted $\hat{\lambda}$. Right: held-out Δ . As K_{neg} increases, the set includes progressively less extreme negatives, so both $\hat{\lambda}$ and Δ decrease.

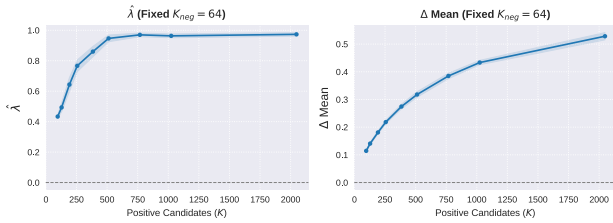


Figure 8. Scaling the negative pool with a fixed hard set ($K_{\text{neg}} = 64$). We fix the number of hard negatives at $K_{\text{neg}}=64$ and vary the pool size K . Left: fitted $\hat{\lambda}$. Right: held-out Δ . As K grows, the hardest negatives become more extreme and $\hat{\lambda}$ increases monotonically toward one.

D.2. Data, candidate sets, and top-1 outcomes

For a frozen checkpoint, we construct an evaluation bank of $N_{\text{batch}} = 32$ batches of size $B = 256$. Each batch contains $2B$ augmented views. For each anchor view $i \in \{1, \dots, 2B\}$, let i^+ denote the paired positive view. The

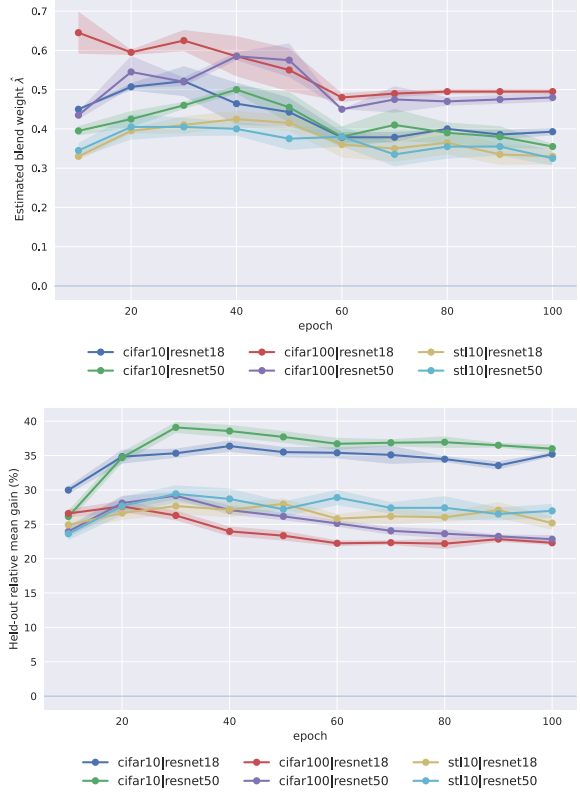


Figure 9. Likelihood based link selection across epochs, datasets and backbones complementary to Figure 4. Top: estimated blend weight $\hat{\lambda}$ (mean across seeds, 95% confidence interval) across epochs. Bottom: held out mean log likelihood gain Δ (nats per anchor; mean across seeds, 95% confidence interval).

candidate set is

$$\mathcal{C}(i) = \{1, \dots, 2B\} \setminus \{i\},$$

and we treat the observed top-1 outcome as $Y_i = i^+$. We compute cosine similarity scores s_{ik} between the normalized embeddings of i and k for all $k \in \mathcal{C}(i)$. This yields a conditional choice dataset $\{(\{s_{ik}\}_{k \in \mathcal{C}(i)}, Y_i = i^+)\}$.

D.3. Models and parameter fitting

We compare the null softmax link (2.7) to the nested blended-coordinate link (2.9)–(2.8). We split batches into held-in and held-out sets (using `train_frac = 0.5`). Parameters are fit by maximizing the held-in top-1 log-likelihood:

$$\hat{\tau}_0 = \arg \max_{\tau_0 > 0} \sum_{i \in \text{held-in}} \log p_0(Y_i = i^+ | \{s_{ik}\}_{k \in \mathcal{C}(i)})$$

and

$$(\hat{\lambda}, \hat{\tau}_1) = \arg \max_{\substack{\lambda \in [0, 1] \\ \tau_1 > 0}} \sum_{i \in \text{held-in}} \log p_1(Y_i = i^+ | \{s_{ik}\}_{k \in \mathcal{C}(i)})$$

Dataset	Backbone	$\hat{\lambda}$	$\bar{\Delta}_{W-0}$	$e^{\bar{\Delta}_{W-0}}$	n_{seeds}
CIFAR-10	ResNet-18	0.393 ± 0.018	0.352 ± 0.006	1.422 ± 0.009	7
CIFAR-10	ResNet-50	0.355 ± 0.011	0.360 ± 0.010	1.433 ± 0.015	10
CIFAR-100	ResNet-18	0.495 ± 0.011	0.223 ± 0.005	1.250 ± 0.007	10
CIFAR-100	ResNet-50	0.480 ± 0.018	0.228 ± 0.010	1.256 ± 0.013	10
STL-10	ResNet-18	0.330 ± 0.035	0.252 ± 0.016	1.286 ± 0.021	10
STL-10	ResNet-50	0.325 ± 0.030	0.270 ± 0.018	1.309 ± 0.023	10

Table 3. Weibull-vs-softmax (nested) link selection across datasets/backbones. All values represent the mean $\pm$ 95% confidence interval half-width across seeds. $\hat{\lambda}$ is the fitted shortfall-mixture weight in the blended tail-coordinate link. $\bar{\Delta}_{W-0}$ is the held-out mean log-likelihood gain (nats/anchor) of the Weibull (shortfall) link over softmax. $e^{\bar{\Delta}_{W-0}}$ is the corresponding *geometric* improvement factor in the probability assigned to the observed top-1 (paired positive).

In our runs we use a grid search with `lam_grid_size = 21` and `tau_grid_size = 21`.

D.4. Held-out effect size

On held-out batches we compute the per-batch mean log-likelihood difference

$$\Delta_b \triangleq \frac{1}{2B} \sum_{i=1}^{2B} \left(\log p_1(Y_i = i^+) - \log p_0(Y_i = i^+) \right), \quad (\text{D.1})$$

and report the held-out mean

$$\bar{\Delta} \triangleq \frac{1}{|\mathcal{B}_{\text{test}}|} \sum_{b \in \mathcal{B}_{\text{test}}} \Delta_b.$$

Because $\bar{\Delta}$ is a mean log-ratio, $\exp(\bar{\Delta})$ equals the ratio of geometric-mean probabilities assigned to the positive under the two links.

D.5. Significance tests and uncertainty quantification

Clustered (batch-level) t -test. Anchors within the same batch share the same negative pool, hence their log-likelihood terms are dependent. We therefore treat each batch as an approximately independent *cluster* and test

$$H_0 : \mathbb{E}[\Delta_b] \leq 0 \quad \text{vs.} \quad H_1 : \mathbb{E}[\Delta_b] > 0$$

using a one-sided t -test over the batch-level statistics $\{\Delta_b\}$.

Nonparametric bootstrap over batches (95% CI). To obtain confidence intervals for $\bar{\Delta}$ without relying on normality, we resample held-out batches with replacement and recompute $\bar{\Delta}$ on each bootstrap replicate. The 2.5% and 97.5% percentiles give a nonparametric 95% CI. (Resampling at the batch level preserves the within-batch dependence structure.)

Parametric bootstrap for a nested likelihood ratio test. Since $\mathcal{H}_0$ is nested in $\mathcal{H}_1$ at the boundary $\lambda = 0$, standard asymptotic χ^2 calibration for the likelihood ratio statistic

may be inaccurate in finite samples. We therefore optionally calibrate the likelihood ratio test using a parametric bootstrap: (i) fit $\hat{\tau}_0$ on held-in data under $\mathcal{H}_0$, (ii) simulate synthetic winners Y_i from $p_0(\cdot \mid \hat{\tau}_0)$ while keeping score vectors $\{s_{ik}\}$ fixed, (iii) refit both $\mathcal{H}_0$ and $\mathcal{H}_1$ on the simulated held-in data, compute the simulated LRT statistic, and (iv) repeat to form an empirical null distribution and a bootstrap p -value.

K -fold cross-validation. To reduce sensitivity to a single train/test split, we also run `cv_k = 5` fold cross-validation over batches. For each fold, we fit parameters on the training folds and evaluate $\bar{\Delta}$ on the held-out fold. We report the mean and standard deviation across folds.

D.6. Fréchet ablation

What is being tested. For each anchor i we observe a candidate set $\mathcal{N}_i$ and a *score vector* $\{s_{ij}\}_{j \in \mathcal{N}_i}$ produced by a frozen encoder checkpoint, along with a top-1 outcome (the paired positive i^+). We *do not* treat these scores as latent utilities. Instead, we treat the score vector as the conditioning statistic and compare *link functions* $p(y = i^+ \mid \{s_{ij}\})$ by held-out top-1 log-likelihood. Since cosine similarities satisfy $s_{ij} \in [-1, 1]$, a bounded-endpoint (shortfall) geometry is a natural candidate for the observed statistic, whereas a heavy-tail (Fréchet) geometry is a priori less plausible; we nevertheless include it as an explicit ablation below.

Test statistic. Given two candidate links p_A and p_B , we report the held-out mean log-likelihood gain

$$\bar{\Delta}_{A-B} \triangleq \frac{1}{|\mathcal{D}_{\text{test}}|} \sum_{(i, \mathcal{N}_i) \in \mathcal{D}_{\text{test}}} \left[\log p_A(i^+ \mid \{s_{ij}\}_{j \in \mathcal{N}_i}) - \log p_B(i^+ \mid \{s_{ij}\}_{j \in \mathcal{N}_i}) \right]$$

measured in nats/anchor. We aggregate over multiple random seeds by averaging $\bar{\Delta}$ within each seed and reporting the mean $\pm$ 95% confidence intervals across seeds.

Dataset	Backbone	$\hat{\lambda}_W$	$\bar{\Delta}_{W-0}$	$\bar{\Delta}_{F-0}$	$\bar{\Delta}_{WF-W}$	$\hat{\lambda}_F^{(\text{full})}$
CIFAR-100	ResNet-18	0.490 ± 0.015	0.206 ± 0.002	0.000 ± 0.000	0.000 ± 0.002	0.075 ± 0.039
STL-10	ResNet-18	0.350 ± 0.038	0.253 ± 0.017	0.000 ± 0.000	0.000 ± 0.001	0.305 ± 0.148

Table 4. Fréchet ablation summary. All values represent the mean $\pm$ 95% confidence interval half-width. $\bar{\Delta}_{F-0}$ is the held-out mean gain of a *Fréchet-only* (heavy-tail) link over softmax. $\bar{\Delta}_{WF-W}$ is the incremental gain from adding a Fréchet component on top of the Weibull (shortfall) model. $\hat{\lambda}_F^{(\text{full})}$ is the fitted Fréchet mixture weight in the joint (Weibull+Fréchet) model. The empirical finding is that Fréchet provides no measurable benefit once shortfall geometry is included ($\bar{\Delta}_{WF-W} \approx 0$).

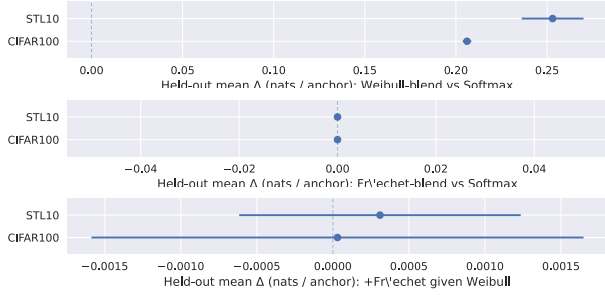


Figure 10. Forest plots of held-out log-likelihood gains (mean $\pm$ 95% CI over seeds). **(top)** $\bar{\Delta}_{W-0}$: Weibull (shortfall) link vs. softmax. **(middle)** $\bar{\Delta}_{F-0}$: Fréchet-only link vs. softmax. **(bottom)** $\bar{\Delta}_{WF-W}$: additional gain from adding Fréchet on top of Weibull. Across datasets/backbones, Weibull yields consistent positive gains, while Fréchet-only and the incremental Fréchet gain given Weibull are near zero.

Conclusion. Across datasets/backbones, including an end-point shortfall coordinate yields a robust increase in held-out top-1 likelihood (Table 3, Figure 10 top). By contrast, a heavy-tail Fréchet component provides no measurable improvement either by itself (middle) or when added on top of the Weibull model (bottom). This supports our focus on a Gumbel + Weibull blend (softmax + shortfall) rather than a three-way mixture including Fréchet.

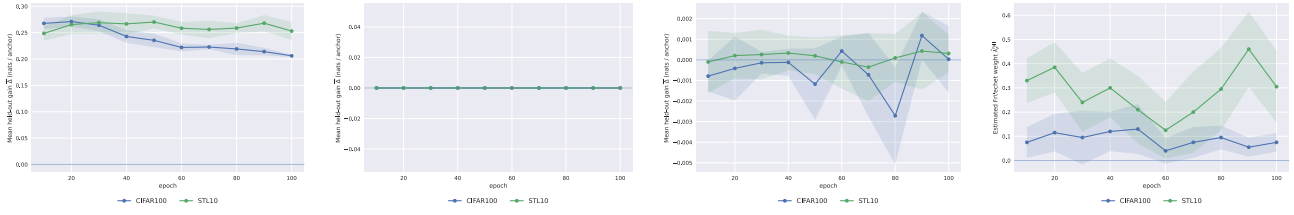


Figure 11. Per-epoch view of model selection quantities (mean $\pm$ 95% CI over seeds). From left to right: (1) Weibull-vs-softmax gain $\bar{\Delta}_{W-0}$ remains positive across epochs; (2) Fréchet-only gain $\bar{\Delta}_{F-0}$ stays near zero; (3) incremental gain $\bar{\Delta}_{WF-W}$ stays near zero, indicating that adding a Fréchet component does not improve predictive top-1 likelihood once shortfall geometry is included; (4) fitted Fréchet weight $\hat{\lambda}_F^{(\text{full})}$ in the joint model.

E. Details of WEINCE

Algorithm 1 gives the main training procedure. Here we provide the derivation of the shortfall coordinate and additional implementation details.

Deriving the endpoint shortfall coordinate. In the Weibull indexing model in Table 1 we write

$$b - X_s = q(s)W, \quad W \geq 0,$$

so that the asymptotic top 1 weights satisfy $p_i \propto q(s_i)^{-\beta}$ by Theorems B.2 and 2.4. In our bounded similarity setting, the relevant endpoint is the score cap, so we take $b = x_F$ (for cosine similarity, $x_F = 1$). It is then natural to assume that $q(s) > 0$ for $s < x_F$ and that $q(s) \rightarrow 0$ as $s \uparrow x_F$, meaning that the utility shortfall vanishes as the score approaches the cap.

Moreover, since the Weibull component of WEINCE targets the extreme regime $s \uparrow x_F$, only the local behavior of q near x_F is identifiable. If q is left differentiable at x_F with $c := -q'(x_F) > 0$, then a first order expansion gives

$$q(s) = c(x_F - s) + o(x_F - s) \quad (s \uparrow x_F).$$

The multiplicative constant c does not affect the induced choice probabilities, since it contributes a common factor $c^{-\beta}$ to $q(s)^{-\beta}$ (equivalently, an additive constant $-\beta \log c$ in the logits) that cancels after normalization. Therefore, as a first-order endpoint model we set

$$q_0(s) = x_F - s \quad (\text{cosine: } q_0(s) = 1 - s).$$

This replacement is asymptotically equivalent in the extreme window, but it is not an exact without-loss-of-generality reduction away from the endpoint. It yields the Weibull shortfall logits $\ell^W(s) = -\beta \log(x_F - s)$ used by WEINCE.

Computational overhead. As reported in the main text, WEINCE adds no forward passes or trainable parameters; in the representative Tiny-ImageNet/ResNet-18 timing run, the additional tail fitting changes the end-to-end step time by less than 0.7%.

Estimating λ_i and the tail index β_i . We estimate λ_i online from the current batch using two signals computed from the negative scores of anchor i . Let $\mathcal{N}(i)$ be the negatives and define shortfalls $\delta_{ij} = 1 - s_{ij}$.

(i) *Cap proximity (hardness).* Define the closest negative shortfall

$$\rho_i \triangleq \min_{j \in \mathcal{N}(i)} \delta_{ij} = 1 - \max_{j \in \mathcal{N}(i)} s_{ij}. \quad (\text{E.1})$$

Small ρ_i indicates that anchor i is operating near the score ceiling.

(ii) *Tail shape evidence and β_i (Weibull versus Gumbel proxy).* Let $\delta_{i,(1)} \leq \delta_{i,(2)} \leq \dots$ be the ordered negative shortfalls, so that $\delta_{i,(1)} = \rho_i$. Using the K_{tail} smallest shortfalls, we form an empirical CDF $\hat{F}(\delta_{i,(k)}) \approx k/(K_{\text{tail}} + 1)$ and fit two one dimensional tail models:

$$\text{(Weibull): } \log \hat{F}(\delta_{i,(k)}) \approx a_i + \beta_i \log \delta_{i,(k)}, \quad (\text{E.2})$$

$$\text{(Gumbel proxy): } \log(-\log \hat{F}(\delta_{i,(k)})) \approx c_i - \kappa_i \log \delta_{i,(k)}. \quad (\text{E.3})$$

From the corresponding residual sums of squares we compute an AIC style score for each fit (up to additive constants) and define

$$\Delta \text{AIC}_i \triangleq \text{AIC}_G(i) - \text{AIC}_W(i), \quad (\text{E.4})$$

so that $\Delta \text{AIC}_i > 0$ favors a Weibull style tail geometry for anchor i . We use the Weibull slope estimate $\hat{\beta}_i$ in the shortfall logits $\ell_{ij}^W = -\hat{\beta}_i \log(1 - s_{ij})$.

(iii) *Soft decision rule.* We convert these signals into a smooth interpolation weight via sigmoids:

$$\lambda_i \triangleq \sigma(\kappa_\rho(\log \rho_0 - \log \rho_i)) \sigma(\kappa_{\text{AIC}}(\Delta \text{AIC}_i - m)), \quad (\text{E.5})$$

where $\sigma(\cdot)$ is the logistic sigmoid, ρ_0 is a pivot (for example, a batch quantile of $\{\rho_i\}$), $m \geq 0$ is an AIC margin, and $\kappa_\rho, \kappa_{\text{AIC}} > 0$ tune the transition sharpness.

Practical details. We compute the shortfall term with clipping for numerical stability:

$$\log(1 - s_{ij}) \leftarrow \log(\varepsilon + 1 - \min(s_{ij}, 1 - \varepsilon))$$

for a small $\varepsilon > 0$. The statistics λ_i and $\hat{\beta}_i$ are computed from the batch similarity matrix and treated as non differentiable per step estimates (that is, gradients are not backpropagated through λ_i or $\hat{\beta}_i$).

Drop in replacement for InfoNCE. WEINCE keeps the SimCLR pipeline unchanged (augmentations, encoder, projection head, batch construction, and the same cross entropy over a $2N \times (2N - 1)$ similarity matrix). The only modification is the logit construction: after computing cosine similarities s_{ij} , we estimate, for each anchor, a tail index $\hat{\beta}_i$ and an interpolation weight λ_i from batch tail statistics of the negative shortfalls $\delta_{ij} = 1 - s_{ij}$, then replace the vanilla logits s_{ij}/τ by the interpolated logits in (3.1). No additional learnable parameters are introduced beyond the SimCLR network, and all tail quantities are recomputed on the fly from the current minibatch similarity matrix and treated as stop gradient statistics.

F. Details for Experiments

Codebase. All experiments use the official SimCLR PyTorch implementation from Spijkervet et al.⁶ as the training/evaluation pipeline. We keep the dataset-specific SimCLR settings from that codebase (augmentations, optimizer, temperature, schedule, etc.) fixed, and change *only* the contrastive loss (InfoNCE vs. WEINCE). The final code will be released upon acceptance.

F.1. Validation split and model selection

A key change relative to earlier drafts is that we use a held-out *validation* split carved out of the labeled training set, and we use this validation split for model selection. Concretely, for each dataset we construct a **stratified 80/20 split** of the labeled training set: 80% is used for training the downstream evaluator (linear probe) / building the kNN feature bank, and 20% is reserved as a validation set. The split is performed independently within each class by shuffling the indices of that class and assigning the first $\lfloor 0.2 n_c \rfloor$ examples to validation and the remainder to training.

What the validation set is used for. When multiple hyperparameter configurations are considered (e.g., multiple WEINCE hyperparameter settings and/or multiple checkpoints), we select the configuration using **validation performance only**. The reported tables show the corresponding **test-set** performance of the configuration chosen on validation (no test-set tuning).

F.2. Backbones and contrastive pretraining

We consider ResNet-18 and ResNet-50 backbones, each followed by the standard SimCLR projection head. For each dataset/backbone/loss configuration we train **four** independent encoders (four different pretraining seeds).

We use the default temperature $\tau = 0.5$ and optimizer settings (Adam with weight decay 10^{-6}) provided in (Spijkervet, 2020), as suggested to be a good temperature choice across batch sizes. We trained encoders for 100 epochs. Following the default configuration of the SimCLR PyTorch codebase, we use a batch size of 256, and a ResNet backbone with ℓ_2 normalization enabled and a projection head of output dimension 64. We use the standard SimCLR two-view augmentation pipeline and train with in-batch negatives (each batch contains $2N$ augmented views).

Losses. The baseline is the standard NT-Xent / InfoNCE objective. For WEINCE, we use the full logit-interpolation loss described in the main text (Equations (3.1) and (3.2) and Algorithm 1), where each anchor has an online-estimated interpolation weight λ_i and tail index $\hat{\beta}_i$ computed from

batch tail statistics. These per-step statistics are treated as stop-gradient quantities (no backpropagation through λ_i or $\hat{\beta}_i$). Aside from the modified logits, the SimCLR pipeline is unchanged.

F.3. Linear downstream classification (linear probe)

We evaluate the pretrained representation quality using a frozen-feature linear probe.

Feature extraction. Given a pretrained checkpoint, we freeze the encoder and compute feature vectors using the *encoder representation* h (the backbone output before the projection head; in our code this is the first output returned by the SimCLR forward pass). Features are extracted using the repository’s evaluation transform (`TransformsSimCLR(...).test_transform`) on: (i) the labeled training split and (ii) the held-out test split.

Train/validation split (20%). From the labeled training split we form a **stratified 80/20** train/validation partition (Section F.1). The linear probe is trained on the 80% partition and evaluated on both the 20% validation partition and the test set.

Linear classifier and optimization. We train a single linear classifier (multinomial logistic regression; a single fully-connected layer) on top of the frozen features with cross-entropy loss. We optimize the probe using Adam with learning rate 3×10^{-4} for 100 epochs (batch size 256). No encoder weights are updated during this stage.

Probe seeds and aggregation. To account for probe-level randomness (initialization, minibatch order, optimizer noise), we train **five** probes per pretrained encoder using evaluation seeds $\{1, 2, 3, 4, 5\}$. This yields $4 \times 5 = 20$ linear-probe runs per dataset/backbone/loss. For each pretrained encoder seed e we average the five probe runs to obtain a per-encoder mean accuracy μ_e (and the corresponding mean validation accuracy). We then treat the four encoder seeds as i.i.d. replicates and report the mean and 95% confidence intervals across encoder seeds (using a t -interval).

Model selection. When selecting among hyperparameter configurations (Section F.1), we use the validation accuracy (averaged over probe seeds for each encoder) to choose the configuration, and we report the corresponding test accuracy.

F.4. kNN evaluation

We also evaluate the frozen representation using a non-parametric k -nearest-neighbor procedure.

⁶<https://github.com/Spijkervet/SimCLR>

Table 5. **k-Nearest Neighbor (kNN) recall (R@k, %)**. We report the mean recall $\pm$ 95% confidence interval halfwidth for the baseline (INFONCE) and our method (WEINCE). Encoders are abbreviated as R18 (ResNet18) and R50 (ResNet50). Best results are highlighted in bold. Table 7 displays the remaining R@50 column in Section F.

Dataset	Enc.	R@1		R@2		R@5		R@10		R@20	
		INFONCE	WEINCE	INFONCE	WEINCE	INFONCE	WEINCE	INFONCE	WEINCE	INFONCE	WEINCE
STL10	R18	68.22 $\pm$ 0.75	70.83 $\pm$ 1.16	79.48 $\pm$ 0.63	80.75 $\pm$ 1.19	90.19 $\pm$ 0.27	91.01 $\pm$ 0.19	95.18 $\pm$ 0.14	95.75 $\pm$ 0.16	97.91 $\pm$ 0.19	98.11 $\pm$ 0.27
	R50	69.66 $\pm$ 0.64	71.75 $\pm$ 0.71	80.39 $\pm$ 0.46	82.03 $\pm$ 0.73	90.65 $\pm$ 0.21	91.31 $\pm$ 0.62	95.42 $\pm$ 0.19	95.64 $\pm$ 0.31	97.97 $\pm$ 0.04	98.18 $\pm$ 0.12
CIFAR10	R18	75.03 $\pm$ 0.22	76.27 $\pm$ 0.40	83.79 $\pm$ 0.10	84.95 $\pm$ 0.33	91.59 $\pm$ 0.24	92.31 $\pm$ 0.21	95.29 $\pm$ 0.05	96.06 $\pm$ 0.29	97.58 $\pm$ 0.13	98.28 $\pm$ 0.11
	R50	74.74 $\pm$ 0.44	77.12 $\pm$ 0.33	83.55 $\pm$ 0.18	85.31 $\pm$ 0.68	91.45 $\pm$ 0.44	92.94 $\pm$ 0.14	95.29 $\pm$ 0.31	96.26 $\pm$ 0.16	97.62 $\pm$ 0.18	98.17 $\pm$ 0.20
CIFAR100	R18	36.98 $\pm$ 0.22	42.94 $\pm$ 0.77	47.42 $\pm$ 0.36	53.29 $\pm$ 0.56	61.72 $\pm$ 0.52	66.19 $\pm$ 0.51	72.18 $\pm$ 0.66	76.11 $\pm$ 0.97	81.45 $\pm$ 0.60	84.41 $\pm$ 0.72
	R50	35.44 $\pm$ 0.40	42.70 $\pm$ 0.79	46.09 $\pm$ 0.71	52.85 $\pm$ 0.68	60.62 $\pm$ 0.42	66.26 $\pm$ 0.39	71.34 $\pm$ 0.43	75.72 $\pm$ 0.43	80.90 $\pm$ 0.70	84.19 $\pm$ 0.37
Tiny-IN	R18	17.29	18.84	25.29	26.64	38.26	39.12	49.87	50.14	61.82	61.90

Table 6. **kNN classifier accuracy at $k = 50$ (%)**. We report the mean top-1 accuracy $\pm$ 95% confidence interval halfwidth for the baseline (INFONCE) and our method (WEINCE). Encoders are abbreviated as R18 (ResNet18) and R50 (ResNet50). Best results are highlighted in bold.

Dataset	Enc.	kNN Acc	
		INFONCE	WEINCE
STL10	R18	73.13 $\pm$ 0.79	74.68 $\pm$ 0.59
	R50	73.93 $\pm$ 0.51	75.72 $\pm$ 0.54
CIFAR10	R18	78.20 $\pm$ 0.39	79.60 $\pm$ 0.25
	R50	77.98 $\pm$ 0.36	79.76 $\pm$ 0.42
CIFAR100	R18	40.73 $\pm$ 0.42	47.38 $\pm$ 0.28
	R50	39.51 $\pm$ 0.53	46.41 $\pm$ 0.24
Tiny-IN	R18	24.41	25.13

Table 7. **k-Nearest Neighbor (kNN) recall (R@50, %)**. We report the mean recall $\pm$ 95% confidence interval halfwidth for the baseline (INFONCE) and our method (WEINCE). Encoders are abbreviated as R18 (ResNet18) and R50 (ResNet50). Best results are highlighted in bold.

Dataset	Enc.	R@50	
		INFONCE	WEINCE
STL10	R18	99.48 $\pm$ 0.05	99.54 $\pm$ 0.08
	R50	99.49 $\pm$ 0.09	99.50 $\pm$ 0.14
CIFAR10	R18	99.06 $\pm$ 0.21	99.47 $\pm$ 0.07
	R50	99.06 $\pm$ 0.05	99.46 $\pm$ 0.07
CIFAR100	R18	90.75 $\pm$ 0.34	92.81 $\pm$ 0.31
	R50	90.35 $\pm$ 0.45	92.43 $\pm$ 0.39
Tiny-IN	R18	76.46	76.70

Feature extraction and similarity. We use the same frozen encoder features h as for linear probing, and ℓ_2 -normalize them. Nearest neighbors are computed using cosine similarity (dot product of normalized features).

kNN bank, queries, and validation split. The kNN *feature bank* is built from the 80% training partition of the labeled training set (Section F.1). We evaluate: (i) on the 20% validation partition (queries = validation, bank = training) and (ii) on the test set (queries = test, bank = training). This ensures that any selection based on validation does not leak test labels.

Metrics. We report **recall at k (R@ k)** of the extracted features for $k \in \{1, 2, 5, 10, 20, 50\}$, where

$$R@k = \frac{1}{|Q|} \sum_{q \in Q} \mathbf{1}\{\text{label}(q) \text{ appears in top-}k \text{ neighbors}\}$$

in Table 5. The main text reports R@1 through R@20 in Table 5; for runs where it was logged, the remaining R@50 column is displayed in Table 7. Our evaluation code additionally computes the standard SimCLR-style temperature-weighted kNN classification accuracy using $k = 50$ and temperature $\tau = 0.1$; this is not required to define R@ k but can be used as an auxiliary score. The results are provided in Table 6.

Split seeds and aggregation. For robustness, we repeat the stratified 80/20 train/validation split for kNN using five different split seeds $\{1, 2, 3, 4, 5\}$, and average the resulting validation/test kNN metrics across these splits. When multiple hyperparameter configurations are compared, selection is performed using the validation kNN metric (with the same averaging over split seeds), and the reported numbers are the test kNN metrics of the selected configuration.